

AN ELEMENTARY TREATISE ON
MICROECONOMICS

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*Being a First Course in Demand, Supply, Cost and
the Behaviour of Markets*

S. SHIR, M. ROUD

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THINK IT THROUGH

2.T1 You can study for one hour or work in a shop and earn Tk 120 an hour. Explain the cost of studying that hour, and say why the answer may differ for another student.

2.T2 Rafiq's week grows to 50 hours. State his new maximum books and parcels, and say whether 600 books and 300 parcels are now attainable.

WHERE PEOPLE GO WRONG

Many students believe that opportunity cost is all the alternatives added together. It is only the best one. They also believe that opportunity cost is the money paid; yet a free evening used for study still has a cost. A third error is to think that scarcity means poverty or rarity, when it describes the meeting of limited resources and unlimited wants for everyone. The last error is to think that a point inside the curve is a good choice because it is attainable. It is attainable, but it wastes resources that could give more of both goods.

IN HISTORY

Lionel Robbins (1898 to 1984) was an English economist who taught at the London School of Economics. In 1932 the publisher Macmillan of London issued his book *An Essay on the Nature and Significance of Economic Science*. Earlier writers had defined economics as the study of wealth or of business. Robbins argued instead that economics studies human behaviour as a relationship between ends and scarce means that have alternative uses.

The new definition changed the question. It did not matter whether an activity concerned money; any choice among uses of limited means had an economic side. The book became one of the most cited works on the nature of the subject, and its idea shapes this chapter: wants are many, means are limited, and choice follows. Not every economist accepted the definition, and some argued that it left out matters such as the study of national output.

A fifth error is to treat a sunk loss as a cost of the next decision. Money already spent and not recoverable cannot be given up again, so only the alternatives still open count when you choose.

REVIEW QUESTIONS

2.1 Define want, and explain why the list of wants as a whole has no end even though one want can be satisfied.

2.2 Explain scarcity in your own words, and give one example for a rich person and one for a poor person.

2.3 A student can spend Saturday studying, working for Tk 400 or resting. Studying is chosen. What is the opportunity cost?

2.4 In one hour a tailor can make 8 shirts or 4 pairs of trousers.

*Four common
misunderstandings, each
corrected.*

*In 1932 an English economist
defined economics as the
study of choice under
scarcity.*

*The definition made scarcity
and choice the centre of the
subject.*

*A fifth error concerns money
already spent.*

*To the one whom Wisdom deliberately chased,
but who was always the faster one in the
chase. RU · MGT-30 · Shoheliya Khanom Rimi*

*Points on, inside and outside
the curve have three different
meanings.*

*The slope of the line shows the
opportunity cost.*

*Real curves usually bow
outward, and the cost rises as
you move along.*

*More resources or better
methods move the whole
curve outward.*

*A whole country can be
pictured with the same curve.*

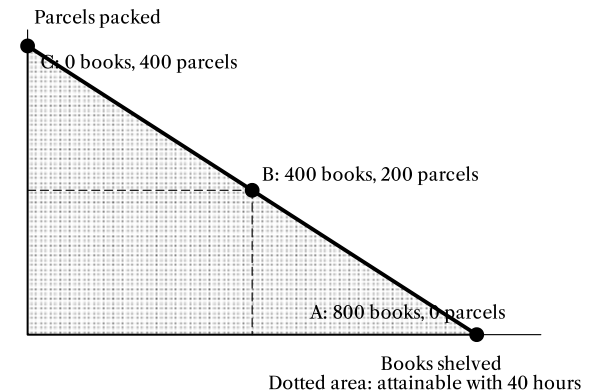


FIG. 2.1. Rafiq's production possibility curve with three points

READING THE CURVE

The curve divides the picture into three regions. A point on the curve is attainable and uses all 40 hours. A point inside the curve is attainable but wastes something. For example, 400 books and 100 parcels needs only $20 + 10 = 30$ hours, so 10 hours are left idle. A point outside the curve cannot be reached with present resources. For example, 600 books and 300 parcels needs $30 + 30 = 60$ hours, which is more than 40.

The curve also shows opportunity cost. Moving along the line from A toward C, each extra parcel gives up 2 books, as we found earlier. The steepness of the line therefore carries the price of one good in terms of the other. A trade-off is visible as a move along the curve: more of one good, less of the other.

Rafiq's curve is a straight line because every hour of his time is equally good at each job. Most real curves bow outward, like a bowl turned on its side. The reason is that resources differ. A farm field suited to rice grows jute poorly, so moving it to jute costs more and more rice. This book returns to the idea when it studies costs of production. For now, remember that a straight line is a simplification chosen for easy arithmetic.

GROWTH SHIFTS THE CURVE

The curve is drawn for fixed resources and fixed methods. If either improves, the curve moves. Suppose Rafiq's week grows from 40 to 50 hours. All shelving now gives 1,000 books and all packing gives 500 parcels, so the whole line moves outward. Faster work has the same effect. A combination that was out of reach becomes possible. Economists call this growth. It does not remove scarcity, because wants rise as well.

The same picture serves a whole country. Imagine a small country that uses all of its farmland, workers and machines to produce food and medicines. A curve shows the largest combinations of the two. A country at a point on the curve can have more medicines only by giving up some food. A country at a point inside the curve

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2.2 OPPORTUNITY COST

WHAT IS GIVEN UP

Return to the traveller. If the traveller walks to the market town, the evening with the old friend is lost. If the traveller walks to the river village, the market sale is lost. In each case the lost evening has a value, even though no money is paid for it. Economists want a word for this loss, so that choices can be compared fairly.

The word is **opportunity cost**, which means the value of the best alternative given up when a choice is made. Notice the words 'best alternative'. When the traveller has three roads to choose from, only the most valuable of the two roads left behind counts. The opportunity cost of an hour spent at a cinema is the best other use of that hour, such as paid work or study, and not all the possible uses added together.

Opportunity cost is not always measured in money. A student who spends Friday evening at a wedding gives up the evening's study. A shopkeeper who keeps the shop open on a holiday gives up a day of rest. Money can measure the cost when the alternative has a money value, but time, effort and enjoyment count equally.

PAUSE AND CHECK 2.P1 You have a free Saturday morning. You can sleep late, study for three hours or earn Tk 300 helping at a shop. If you choose to study, what is the opportunity cost?

A related word describes the act itself. A **trade-off** means giving up some of one thing in order to have more of another. A farmer who plants more jute plants less rice. A student who studies an extra hour gives up an hour of sleep. Opportunity cost measures the size of what is given up; the trade-off is the exchange itself.

Opportunity cost looks different from each viewpoint. For the owner of a shop, the cost of opening on a holiday is the rest and family time given up. For an employee, the cost of an extra shift is the evening at home. For a consumer, the cost of buying a book is the other goods the same money could have bought. In each case the question is the same: what is the best use that is left behind?

There is an exception. If the best alternative has no value, the opportunity cost is zero. A shop table that stands empty overnight and has no other use costs nothing to use for a night's storage. Such cases are rare, because most resources have some other use, and economists therefore look for the alternative before they call anything free.

You should now be able to say that the opportunity cost of a choice is the best alternative left behind, and that a trade-off is the exchange of some of one thing for more of another.

MEASURING OPPORTUNITY COST

When the choice involves producing goods, opportunity cost can be measured exactly. In words: the opportunity cost of one unit of

GO SLOWLY HERE.

Every choice has a price in what is left behind, even when no money changes hands.

Cost. Word in use. In daily life, the price paid for a good. In this book, price is what the consumer pays, and cost is what is given up or spent to make or obtain something.

Opportunity cost. The value of the best alternative given up when a choice is made.

Opportunity cost is the value of the best alternative that is given up.

Opportunity cost often has no price tag, yet it is real.

Trade-off. Giving up some of one thing in order to have more of another.

A trade-off is giving up some of one thing to gain more of another.

The owner, the employee and the consumer each see the cost differently.

When the alternative has no value, the opportunity cost is zero.

You should now be able to say what the opportunity cost of any choice is.

Opportunity cost can be turned into a number when outputs are known.

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches An Elementary Treatise on Microeconomics one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

YOU WILL NEED

Chapter 1, 'What Economics Is', section 1.1 'The Meaning and Scope of Economics', for the meaning of economics; Chapter 1, section 1.4 'Economic Laws', for laws and the same-conditions rule; and Appendix A1, 'Maths You Will Use', for ratios and simple graphs.

A traveller walks along a country road at the end of an afternoon. Ahead, the road divides in two. A signpost stands on each branch. One road leads to a market town, where the traveller could sell goods and buy supplies. The other leads to a river village, where an old friend waits and a good meal is promised. Night is coming, and there is strength for only one more journey today.

The traveller does not lack roads, and does not lack good reasons. Both roads are worth taking. What the traveller lacks is time, because the evening allows only one walk. Whichever road is chosen, something valuable is left behind. No clever planning removes that loss. It can only be understood, measured and weighed against the gain.

The same pattern appears everywhere. A family with Tk 20,000 to spend cannot buy everything on its list. A shopkeeper with one free afternoon cannot both count stock and visit a supplier. A government with one budget cannot build every road and school it would like. This chapter gives the pattern a name and a measure. Section 2.1 explains why choice cannot be avoided. Section 2.2 shows how to measure what choice gives up. Section 2.3 draws the idea as a curve.

* * *

2.1 WANTS AND SCARCITY

WANTS

Begin with your own week. This morning you wished for a good breakfast, and you had it. By noon you wished for a new pair of shoes. By evening you wished for a holiday by the sea. Each wish was met or set aside, and each was soon replaced by another. You never reached a day with no wish left. This simple observation is the first step in economics.

Economists give such wishes a name. A **want** means something a person would like to have, however little or much the person can pay for it. A bicycle is a want. So are a meal, a university degree, an afternoon of rest and a quiet street. The definition does not ask whether the want is wise or sensible. A want need not be a physical good; it may be a service, such as a haircut, or a condition, such as free time.

Wants differ from person to person and from place to place. A farmer in Mymensingh, a banker in London and a student in Dhaka will list different wants. Wants also change with age and with income. Yet one feature is shared. When one want is satisfied

A traveller must choose one road, and the other road is lost.

The traveller's trouble is not a lack of roads but a lack of time.

The same pattern appears in every shop, household and country.

GO SLOWLY HERE.

Everyday life shows how one wish leads to the next.

Want. Something a person would like to have, however little or much the person can pay for it.

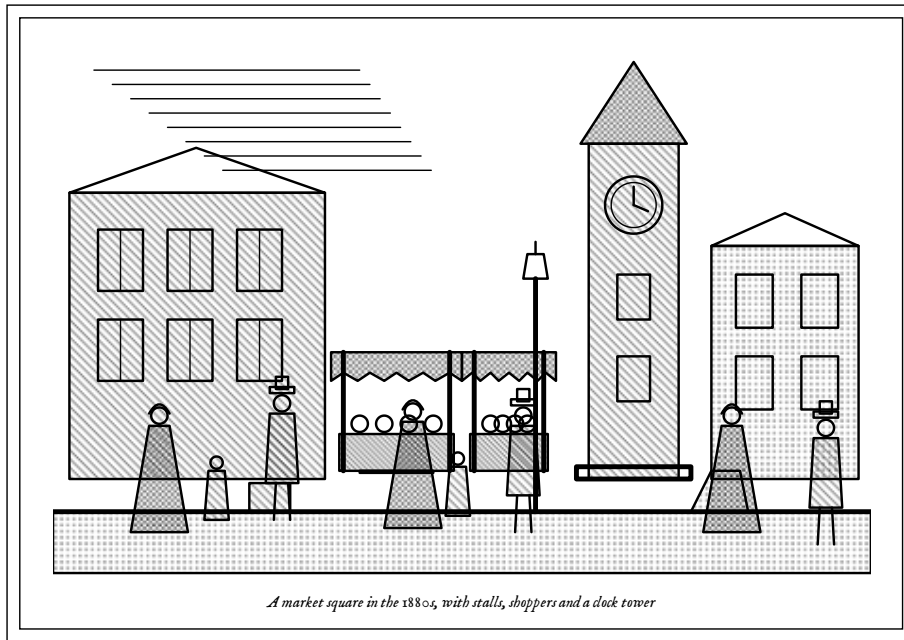
A want is anything a person would like to have.

Wants differ among people, but wishing never ends.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

REVIEW QUESTIONS

- 1.1 Classify each statement as positive or normative, and give a reason: (a) Meghna and Sons had 7 staff in 2025. (b) Bookshops ought to pay their staff more. (c) The shop's profit in 2025 was Tk 750,000. (d) Fewer people should buy books online.
- 1.2 Separate each question into micro or macro: (a) How should one bookshop choose its staff? (b) Why did prices across the country rise? (c) How does a consumer choose between two goods? (d) What decides total employment in the country?
- 1.3 In your own words, explain why a student with Tk 100 and wants worth Tk 150 faces an economic problem.
- 1.4 Name the four activities in the scope of economics and give one example of each from a bookshop.
- 1.5 What does *ceteris paribus* mean, and what is one difference between an economic law and a law of nature?



CHAPTER I

What Economics Is

Why can no one have everything they want?



This chapter introduces economics as the study of choice when wants are many and means are few. It separates the study of single shops from the study of a whole country. It teaches you to tell a statement of fact from a statement of opinion, and to see why every economic law carries a condition. It also sets the habits this book uses in every chapter.

between building a school and building a road faces the same problem on a larger scale.

IN SHORT

Economics is about choice. Wants are many, resources are limited, and so every person, firm and government must choose.

Figure 1 shows the idea in a picture. Read it from left to right. The tall bar on the left stands for wants, and the shorter bar on the right stands for the means to satisfy them. The dashed line marks the top of the means. The distance between the dashed line and the top of the wants is the gap that cannot be filled, and the word *choice* sits between the bars because the gap forces it.

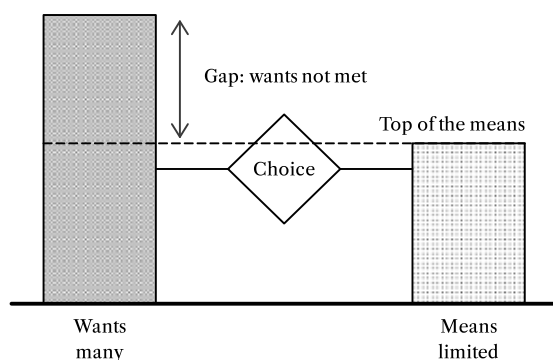


FIG. 1.1. Economics as choice among wants and limited means

The *scope* of a subject is the range of things it studies. Economists usually group the activities they study into four. Consumption is the use of goods to satisfy wants, as when a family eats bread. Production is the making of goods, as when a printer makes books. Exchange is the buying and selling by which goods pass from one person to another. Distribution is the sharing of the income from production among those who helped to make the goods, such as workers and owners. The four are linked: nothing is consumed until it is produced and exchanged, and the income shared in distribution pays for the next round of consumption.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna. In 2025 its sales were Tk 14,400,000, its profit was Tk 750,000, and it had 7 staff. The shop buys books from printers, which is exchange. Readers take the books home to read them, which is consumption. The shop's income pays its staff and its owner, which is distribution. Printers who make the books are engaged in production.

Is economics a science? A science is organised knowledge, built by gathering facts and testing explanations against them. Economics does this: it collects facts about prices, output and choices, and it

Read the first figure as two bars with a gap between them.

Economics covers four connected activities.

The running case shows all four activities in one small shop.

Economics is a science that explains and an art that advises.

The situation lists five items drawn from the shop's case file.

The question asks for two sorting decisions per item.

Items (a) and (e) can be checked against records, so they are positive.

Item (b) states a value, and item (c) asks about a total.

Item (d) is about one firm, so it is a micro question.

The conclusions are collected in a table.

A check confirms each decision with the test from the chapter.

The chapter's main ideas, in a few lines.

or macroeconomics. It is an exercise in reasoning, and the steps are shown in full.

Case file, illustrative. Situation: five items are taken from the file of Meghna and Sons. (a) "The shop sold 41,000 copies in 2025." (b) "Textbooks should be cheaper." (c) "What decides the total number of jobs in the whole country?" (d) "How did the shop's owner choose the number of staff?" (e) "The shop's profit in 2025 was Tk 750,000."

Question: for each statement, decide whether it is positive or normative. For each item, decide whether its subject is micro or macro. A question that asks for information is not a statement, so for items (c) and (d) only the second decision applies.

Reasoning, part one. Item (a) can be checked against the shop's sales books, so it is positive. Its subject is one shop, so it is micro. Item (e) is also a statement of fact: the shop's accounts show its profit, and the figure is Tk 750,000. It is positive, and it is micro, because it describes one firm.

Reasoning, part two. Item (b) says what ought to be, and no record can settle it, so it is normative. It concerns the price of one kind of good, so it is a micro matter. Item (c) asks about the total number of jobs in the whole country, so its subject is macro. It is a question, and so it is neither positive nor normative.

Reasoning, part three. Item (d) asks how one owner chose the number of staff. The unit is one firm, so the question is micro. The answer would be a matter of fact, such as the shop's sales and wages, though the item itself is a question.

Conclusion: the table below gives the result. Read each row across, from the item to its two sorting decisions.

SORTING THE FIVE ITEMS

ITEM	POSITIVE OR NORMATIVE	MICRO OR MACRO
(a) 41,000 copies sold	Positive	Micro
(b) Textbooks should be cheaper	Normative	Micro
(c) Total jobs in the country	Not a statement	Macro
(d) The owner's choice of staff	Not a statement	Micro
(e) Profit of Tk 750,000	Positive	Micro

Check: for each statement, ask whether records could show it to be true or false. Items (a) and (e) pass, and item (b) fails. For each subject, ask whether the unit is one firm or a total. Items (a), (b), (d) and (e) concern single units, and item (c) concerns a total. The decisions agree.

* * *

SUMMARY

Economics is the study of how people, firms and governments choose to use limited resources to satisfy wants. It has two

by hatching. Under each picture, the label names what that view measures. Read the figure as a change of distance, not a change of subject.

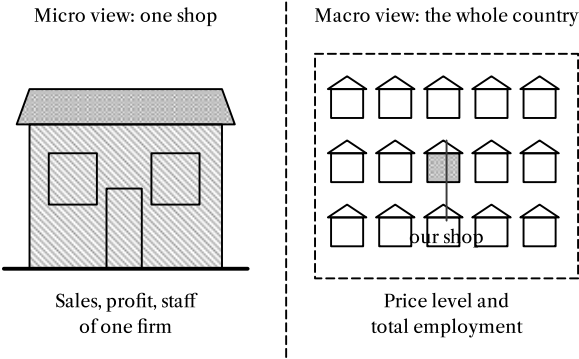


FIG. 1.2. Micro and macro viewpoints on one shop

The table below compares the two branches. Read it across. Each row names one feature, then gives the micro answer in the middle column and the macro answer in the right column.

MICRO AND MACRO COMPARED		
FEATURE	MICROECONOMICS	MACROECONOMICS
Unit studied	One consumer, one firm or one market	The whole economy
Typical question	How does one firm choose its price?	Why did the price level of the country change?
Case file example	Sales of Tk 14,400,000 and 7 staff at one shop	The price level and total employment of the country
Main concern	Single prices and quantities	Totals and averages

The branches differ in the size of the unit, yet they are linked. Total employment in the country is the sum of the jobs in every firm, so macro totals come from micro choices. The link runs the other way as well. When the price level of the country rises, a shop pays more for its supplies, and that changes the shop's choices. One exception needs care. Size alone does not decide the branch: a very large firm is still one firm, so a question about it belongs to microeconomics.

What is the scope of microeconomics? It covers how a consumer chooses among goods, how a firm chooses what to make and how to make it, how prices form where buyers and sellers meet, and how the people who supply work, land and money are paid. This book teaches microeconomics. Macroeconomics appears here only so that you can see where the boundary lies.

Before moving on. Microeconomics studies single consumers and firms. Macroeconomics studies the whole economy. The branches

A table sets the two branches side by side, row by row.

The branches differ in size but are linked in both directions.

This book teaches microeconomics, so its scope is the choices of single units.

Recap of the section in a few lines.

1.3 POSITIVE AND NORMATIVE STATEMENTS

Two friends argue about textbooks, and only one dispute can be settled by checking.

Positive economics deals with statements that facts can test.

Positive economics. Statements about what is, which can be tested against facts.

The shop's own record gives a clean example of a positive statement.

Normative economics deals with what ought to be, which depends on values.

Normative economics. Statements about what ought to be, which depend on values and cannot be tested by facts alone.

One question separates the two kinds of statement.

A table sets the two kinds of statement side by side.

Two friends sit in the shop and discuss books. The first says, "The shop sold 41,000 copies in 2025." The second says, "Textbooks should be cheaper." Both sound like ordinary remarks, but they are different in kind. Could a person settle either one by looking at records? Think about this before you read on.

The first remark is an example of **positive economics**. Positive economics means statements about what is, which can be tested against facts. To test the remark, a person counts the copies recorded in the shop's sales books. The records will show the statement to be true or false. A positive statement may be false, because positive means testable, not correct. If the records show 39,000 copies, the statement fails the test but is still positive.

Case file, illustrative. Statement: "The shop sold 41,000 copies in 2025." This is positive. The shop's sales books can confirm it or reject it, and no opinion is needed to decide.

The second remark is an example of **normative economics**. Normative economics means statements about what ought to be, which depend on values and cannot be tested by facts alone. A value is a belief about what is good or fair. The remark "Textbooks should be cheaper" gives no fact to check. A student who wants low prices and a bookseller who wants a fair return can agree on every fact and still disagree on this point. Words such as should, ought, fair and better often signal a normative statement.

To tell them apart, ask one question: could records or measurement show this statement to be true or false? If the answer is yes, the statement is positive. If the answer depends on what we think is right, the statement is normative. Many statements mix the two. "Textbook prices are too high because paper costs more" has a positive part, the claim about cost, and a normative part, the words too high. Separate the parts and test each one on its own.

The table below compares the two kinds of statement. Read each row across, from the feature on the left to the answers for positive and normative statements.

POSITIVE AND NORMATIVE STATEMENTS COMPARED		
FEATURE	POSITIVE	NORMATIVE
Speaks about	What is	What ought to be
Can facts test it?	Yes	Not by facts alone
Case file example	The shop sold 41,000 copies in 2025	Textbooks should be cheaper
Signal words	Is, was, rose, fell	Should, ought, fair, better

PAUSE AND CHECK 1.P1 Classify each statement as positive or normative, and give a reason: (a) The shop employs 7 staff. (b) The shop should employ an eighth person.

Before moving on. A positive statement says what is and can be tested against facts. A normative statement says what ought to be

You may know the word from daily life, where a *utility* is a public service such as the supply of water or electricity. In this book the word always has its economic sense: satisfaction. When a company that supplies water appears in an example, we call it a water company and keep the word utility for the feeling.

Now suppose the diner drinks three cups. Each cup gave her some satisfaction. If we add the satisfaction from the first cup, the second cup and the third cup, we reach **total utility**. Total utility means the whole satisfaction from all units of a good consumed. For the diner, it is everything she enjoyed from all three cups together. Total utility answers one question: how much satisfaction did she gain altogether?

Now ask a different question: how much did the third cup add? The answer is the **marginal utility** of the third cup. Marginal utility means the extra satisfaction from consuming one more unit of a good. If the diner had two cups and gains a little from a third, that little gain is the marginal utility of the third cup. A staircase gives a useful picture. Total utility is how high you stand after climbing some steps. Marginal utility is the height of the last step you climbed. They are two different measures: the height so far, and the height of one step.

The word *marginal* needs care for the same reason. A margin is the edge of a page. In economics the margin is the edge of what you already have, and marginal means what one more unit adds at that edge. The word does not mean small or unimportant. The marginal utility of a cup may be large, as it is for the first cup after a long walk.

PAUSE AND CHECK 3.P1 The diner drinks one cup, then a second cup that pleases her a little. After the second cup, is her total satisfaction higher or lower than after the first? Give a reason.

MEASURING SATISFACTION

Can satisfaction be counted? Nobody owns a machine that reads how happy a cup of tea makes a person. William Stanley Jevons, an English economist who wrote in the 1870s, found it useful to imagine such a machine.¹ Economists therefore agree to count satisfaction in an imaginary unit.

The unit is the **util**. A util means an imaginary unit used to count satisfaction in economic examples. If the first cup gives 10 utils and the second gives 8 utils, we say that the first cup pleased the diner more. The numbers are not real measurements. They are a tool that lets us reason exactly about more and less.

The table below shows the diner's tea in utils. The numbers are imaginary and chosen to be small. Each row is one cup. The middle column gives the marginal utility of that cup, and the last column gives the **total** utility after that cup.

¹ Recall: opportunity cost, Chapter 2, 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost'. We use it later in this chapter, when we ask what a buyer forgoes.

Utility. Word in use. In daily life, a utility is a public service such as water or electricity. In economics, utility is the satisfaction a good gives.

The everyday word and the economic word differ.

Total utility. The whole satisfaction from all units of a good consumed.

Adding the satisfaction from every unit gives total utility.

GO SLOWLY HERE.

Marginal utility. The extra satisfaction from consuming one more unit of a good.

Marginal utility looks only at the last unit, not at the whole.

Marginal. Word in use. In daily life, marginal often means small or at the edge, as in a marginal improvement. In economics, marginal means extra: the change caused by one more unit.

In economics, marginal means extra.

Satisfaction cannot be seen, so economists count it with an imaginary unit.

Util. An imaginary unit used to count satisfaction in economic examples.

The imaginary unit is the util.

A small table shows five cups of tea, with the extra and the whole counted in utils.

YOU WILL NEED

Chapter 3 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'; Chapter 4 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; Appendix A1 'Maths You Will Use'.

On a market day in a small town, a woman passes a stall that sells water pots. Her old pot has cracked, so she needs a new one. In her own mind she has fixed a limit: she would pay Tk 100 for a good pot, and not one taka more. The stallholder asks Tk 60. She pays at once, takes the pot and walks home.

Now examine the exchange as an accountant would. The woman handed over Tk 60 and received a pot. The stallholder received Tk 60 and handed over a pot. Each side has a record, and the records agree. Yet the woman feels that she has done well, and she is right. The pot was worth Tk 100 to her, and she paid only Tk 60. Forty taka of satisfaction appears on no receipt, in no ledger and in no bank account.

Here is the puzzle. If trade is a fair exchange, how can one side gain something that nobody pays for? And if she gains, can we measure the gain? Many students find the question strange at first, because we are used to thinking that the price tells us what a good is worth. This chapter shows that the price tells us only part of the story. We begin with what consumers do. Then we measure what they gain.

* * *

6.1 CONSUMPTION AND CHOICE

Think about your own day. You eat breakfast, travel by bus, read a message on your phone and drink water. Each act uses something that other people made or provided, and each act satisfies a want. Economics gives this ordinary activity a name, and the name lets us study it with care.

The word **consumption** means the use of goods and services by people to satisfy their wants. A student who eats a meal is engaged in consumption. So is a family that uses electricity to light a room. Here a *good* is a physical thing, such as a notebook, and a *service* is work done for a person, such as a haircut.

Two points prevent confusion. First, buying is not the same as consuming. A consumer may buy a notebook on Monday and use it over many weeks, filling one page after another. The consumption happens during the use, not at the moment of payment. A haircut is different, because the service is used as it is given.

Second, consumption is the final purpose of production. A firm that makes cloth does not consume the cloth. A tailor buys it to make shirts, and only the person who wears a shirt consumes it. A good used to make other goods is therefore not consumption in this sense.

A woman pays Tk 60 for a water pot that she values at Tk 100.

The sale is recorded as Tk 60, yet she feels richer by Tk 40.

The puzzle: how can a gain exist that nobody pays for, and can it be measured?

Everyday acts such as eating or travelling use things that others provide.

Consumption. The use of goods and services by people to satisfy their wants.

Consumption is the use of goods and services to satisfy wants.

Buying is not consuming: the use may come later than the payment.

Only the final user consumes; a firm that uses a good to make other goods does not.

THE LAW IN SHORT

Keep consuming one good, and the extra satisfaction from each new unit becomes smaller.

This point is the one that confuses students most. When marginal utility falls from 10 to 8 to 5, the total still rises, from 10 to 18 to 23. A smaller step is still a step upward. The total utility rises as long as the marginal utility is above zero. At the fifth cup the marginal utility is zero, so the total stops rising and stays at 25. If a sixth cup were forced upon the diner, it would give her discomfort, and the marginal utility would fall below zero.

WHY THE LAW HOLDS

Why should the extra satisfaction shrink? Begin with the bridge from daily life. A person who is very thirsty has a strong want for water. The first glass meets the strongest part of that want, and the relief is great. After the glass, the want is smaller. The second glass meets what remains, so the relief is smaller. Each unit uses up part of a want that has a limit, and a smaller want leaves less satisfaction to gain.

Return to the puzzle at the start of the chapter. If marginal utility did not fall, the diner would drink cup after cup, and no shop would ever need to attract new customers, because each customer would buy forever. Because marginal utility falls, every buyer reaches a point where one more unit is not worth having. When the unit is not free, a buyer must also compare the satisfaction gained with what is forgone, which is the opportunity cost of the purchase. A later chapter joins these two ideas.

Look at the same fact from two viewpoints. For the consumer, it is a feeling: the fourth cup is not worth much. For the owner of a shop such as Meghna and Sons, it is a pattern in sales: a customer buys the first notebook readily, hesitates over the second and rarely wants a long row of identical ones. The owner cannot see the utils, but the owner can see the customer stop. The law gives a reason for what the owner observes. An assistant who serves the customers sees the same fact from a third position: a customer who comes in for one book rarely leaves with ten, and the assistant learns to expect it.

THE SCHEDULE AND THE CURVE

A table that lists the utility of each unit is called a schedule, and the same numbers drawn as a line are called a curve. We now use the shop's own example. The next paragraph is a record of a case, and its figures are fixed.

Case file, illustrative. A student buys exercise books at Meghna and Sons. The marginal utility of each book, in utils, is: 1st book 40, 2nd book 30, 3rd book 20, 4th book 10, 5th book 0. The total utility after each book is 40, 70, 90, 100 and 100.

The law is about the extra, so the total can still rise while the extra falls.

A want is limited, and each unit satisfies part of it.

The law helps explain why buyers do not buy without end.

The same fact looks different from the consumer's side and the seller's side.

A table of utility for one person is a schedule; a line through the numbers is a curve.

The case file records the student's utility for five books.

5.6 A rise of 5% in the price of tea lowers the quantity of sugar demanded by 3%. Calculate cross elasticity and say what the sign shows.

5.7 Give one reason why demand for a brand of tea is more elastic than demand for tea in general.

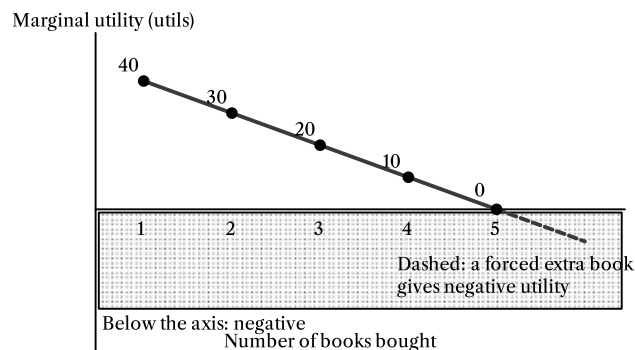


FIG. 3.2. Marginal utility falls with each book and reaches zero

The second chart shows the extra from each book instead of the whole. Its points go down: 40, 30, 20, 10, 0. A line through them slopes downward from left to right, and that downward slope is the law in picture form. The line reaches the horizontal axis at the fifth book, where marginal utility is zero. The dashed part below the axis shows what would happen if a further book were forced upon the student: marginal utility would be negative, and total utility would fall.

Before moving on. The law says the extra satisfaction from each unit falls. The marginal column goes down while the total column goes up and then stops. At the point where marginal utility is zero, total utility is at its highest. You should now be able to say why the total can rise while the extra falls.

3.3 LIMITS OF THE LAW

The law holds when certain conditions are met. The units must be alike in size and quality, as cups of the same tea are. The consumer's tastes must stay the same during the time of consumption. The consumption must follow without long gaps, because after a rest the want returns and the first unit pleases again. Economists say these conditions together as: other things remain equal.

Some cases look like exceptions. A collector of stamps may value each new stamp more as the set nears completion. Yet the unit that matters is then the complete set, and the law holds for sets. A person who is addicted to a drug may crave more each time, but this is a harm and not a normal want. Treat such cases as limits of the law, not as proofs that it fails. A third case is a good that is useful only in a set, such as a left shoe: the first shoe alone adds little, and the second adds much. Here again the sensible unit is the pair.

Before moving on. The law needs alike units, steady tastes and no long gaps. Apparent exceptions usually change the unit.

The second chart shows the same facts as the extra from each book.

The two charts are the same story told twice.

A law holds only under stated conditions.

Apparent exceptions usually disappear when the unit is defined with care.

The section closes with its conditions in two lines.

Cross elasticity. The percentage change in quantity demanded of one good divided by the percentage change in the price of another.

Cross elasticity links the quantity of one good to the price of another.

A 10% dearer rival notebook lifts the shop's notebook by 4%.

Before moving on: income elasticity and cross elasticity use the same ratio.

Cross elasticity means the percentage change in quantity demanded of one good divided by the percentage change in the price of another. A positive result means the goods are substitutes, since a dearer rival sends consumers to the good. A negative result means the goods are used together, so a dearer partner reduces the use of both.

Case file, illustrative. A rise of 10% in the price of a rival notebook raises the quantity demanded of the shop's notebook by 4%. Cross elasticity = $4\% \div 10\% = 0.4$.

Before moving on. Each measure divides a percentage change in quantity demanded by a percentage change in its cause. Income elasticity uses consumers' income; cross elasticity uses the price of another good. Their signs carry meaning.

* * *

WORKED EXAMPLES: CHANGING THE SETTING

The first two examples are above: the tea stall and the notebook. The next two reshape the problem. Both follow the same order: Given, Find, Formula, Steps, Answer, Check.

Given: a rice seller raises the price from Tk 80 to Tk 100 a kilogram. At Tk 80, households buy 30 kg a week; at Tk 100, they buy 100 kg a month. Take a month as four weeks. *Find:* the price elasticity of demand. *Formula:* $E = \frac{\% \Delta Q}{\% \Delta P}$, with midpoint percentages. The two quantities are in different periods, and this is the slip to avoid.

Steps. Convert 100 kg a month to $100 \div 4 = 25$ kg a week. Quantity: change = $25 - 30 = -5$; average = 27.5; $\% \Delta Q = -5 \div 27.5 \times 100 = -18.18\%$. Price: change = $+20$; average = 90; $\% \Delta P = 20 \div 90 \times 100 = +22.22\%$. Divide: $18.18 \div 22.22 = 0.82$. *Answer:* $E = 0.82$, so demand is inelastic. *Check:* total revenue per week is Tk $80 \times 30 =$ Tk 2,400 before and Tk $100 \times 25 =$ Tk 2,500 after. A price rise with inelastic demand raises revenue, and it did.

Given: a bus company's schedule shows 800 passengers a day at a fare of Tk 10, and 700 at a fare of Tk 12. The company asks whether raising the fare from Tk 10 to Tk 12 will increase its total revenue. *Find:* E and the change in total revenue. *Steps.* Quantity: change = -100 ; average = 750; $\% \Delta Q = -13.33\%$. Price: change = $+2$; average = 11; $\% \Delta P = +18.18\%$. $E = 13.33 \div 18.18 = 0.73$. Revenue: Tk $10 \times 800 =$ Tk 8,000 before, Tk $12 \times 700 =$ Tk 8,400 after. *Answer:* demand is inelastic and revenue rises by Tk 400. *Check:* the elasticity test and the direct multiplication agree, so the working is sound.

* * *

SUMMARY

Price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price. Found with

The chapter's main results in a few lines.

Error 4: "Ten utils for me equal ten utils for you." This is wrong. A util is an imaginary unit for one person's own comparison, and no common scale links two people.

IN HISTORY

In 1854, a Prussian civil servant named Hermann Heinrich Gossen published a book in Germany that stated the idea of diminishing satisfaction and drew its curves. Few people bought it, and Gossen died in 1858, largely unknown. In 1871, William Stanley Jevons in England published *The Theory of Political Economy*, and in the same year Carl Menger in Austria published *Grundsätze der Volkswirtschaftslehre*. In 1874, Léon Walras, who taught at Lausanne in Switzerland, published the first part of *Éléments d'économie politique pure*.

The three writers did not know of one another's work when they began. Historians call these years from 1871 to 1874 the marginal revolution, because each author built a theory of value on marginal utility. Jevons learned of Gossen's book after publishing his own, and in a later edition he acknowledged that Gossen had come first.

The story carries a lesson for a student. Gossen stated the idea first and was ignored, while the three later writers were read and remembered. An idea becomes part of a science only when other people understand it, test it and use it. The law you have learned passed that test, and today it is the starting point of the theory of how buyers choose.

REVIEW QUESTIONS

- 3.1 Define utility, and give one example from your own day.
- 3.2 Complete the table. Marginal utilities for four units are 50, 35, 20 and 5 utils. Find the total utility after each unit.
- 3.3 Total utilities for five units are 20, 36, 48, 54 and 56 utils. Find the marginal utility of each unit.
- 3.4 State the law of diminishing marginal utility, and name one condition under which it holds.
- 3.5 In the case table, what is the marginal utility of the 5th exercise book? What is the opportunity cost of buying that book, in words, if the student could spend the money on another good?
- 3.6 A student says the total falls after the third cup because the marginal utility falls. Explain the mistake.

Utils are a tool, not a measurement of the world.

Four writers found the rule, and three of them did so within four years.

The revolution was the work of independent minds in different countries.

The history teaches that a good idea can wait for its readers.

Six questions that check the chapter's ideas.

The three cases are compared in one table.

Total revenue. Price multiplied by quantity sold. *Total revenue is price times quantity, the money a firm receives from sales.*

The total revenue test links the size of the result to the change in revenue.

The notebook's total revenue rises when its price falls.

Along one schedule, revenue rises and then falls as the price keeps falling.

The table below compares the three cases. Read each row from left to right: the first column names the case, the second gives the size of the result, the third says which percentage is larger, and the fourth anticipates the test described next.

THE THREE CASES OF PRICE ELASTICITY OF DEMAND

CASE	SIZE OF RESULT	LARGER PERCENTAGE	EFFECT OF A PRICE CUT ON MONEY RECEIVED
Elastic demand	Above 1	Quantity	Rises
Unit elastic demand	Equal to 1	Neither	Unchanged
Inelastic demand	Below 1	Price	Falls

Why does the result matter to a firm? Because of **total revenue**, which means price multiplied by quantity sold. If a stall sells 10 cups at Tk 5, its total revenue is Tk 50. A price cut lowers the price paid for every unit but raises the number of units sold. Which effect wins depends on the elasticity.

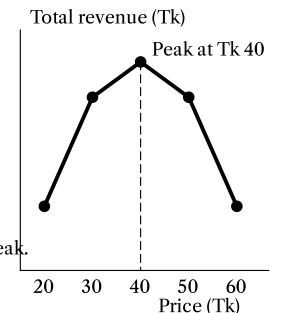
The total revenue test says: if demand is elastic, a fall in price raises total revenue; if demand is unit elastic, total revenue does not change; if demand is inelastic, a fall in price lowers total revenue. A rise in price gives the opposite results. The notebook confirms the first rule, as the next record shows.

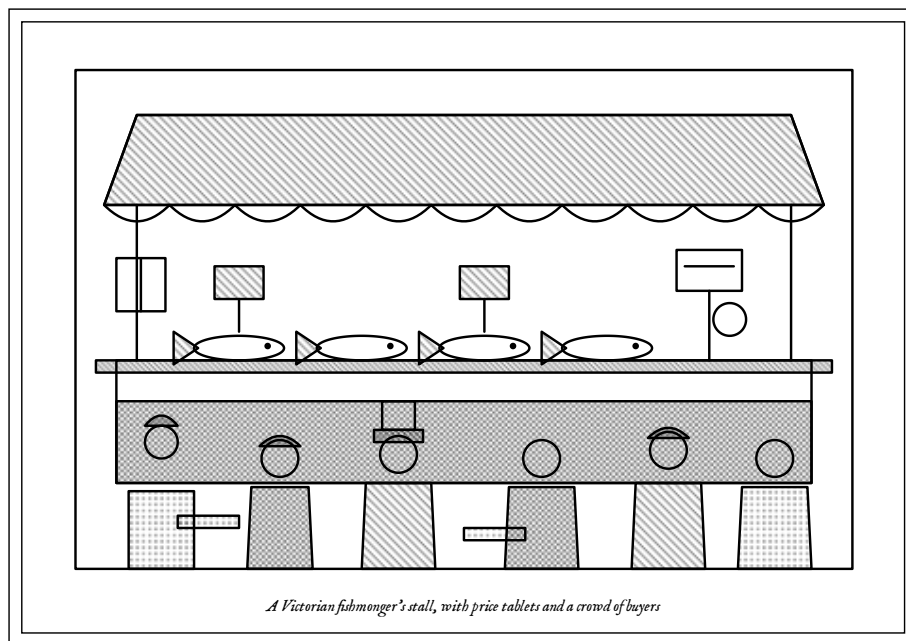
Case file, illustrative. When the notebook's price falls from Tk 50 to Tk 40, total revenue rises from Tk 1,500 (Tk 50 × 30) to Tk 1,600 (Tk 40 × 40). The price elasticity of demand over this fall is 1.29, so demand is elastic.

To read the next figure, look at the table on its left and the graph on its right. In the table, each row gives a price, the quantity demanded from the rule quantity = 80 - price, and total revenue. In the graph, the horizontal axis shows price, the vertical axis shows total revenue, and each point is one row of the table. Total revenue climbs as the price falls from Tk 60 to Tk 40, and then drops. The peak lies where demand changes from elastic to inelastic.

Price	Quantity	Total revenue
Tk 60	20	Tk 1,200
Tk 50	30	Tk 1,500 ^{1,600}
Tk 40	40	Tk 1,600
Tk 30	50	Tk 1,500
Tk 20	60	Tk 1,200 _{1,200}

Quantity = 80 - price. Boxed row: the peak.





CHAPTER IV

Demand and the Demand Curve

Why does a crowd gather round a stall when the price tablet changes?



This chapter builds the central diagram of the book. It explains what buyers are willing and able to buy at each price, why a lower price brings a larger quantity, how to draw the demand curve, and what moves or shifts it. It ends with related goods and a few unusual cases.

demand and $\% \Delta P$ for the percentage change in price. The Greek capital letter Δ (delta) means 'change in'.

IN SHORT

Price elasticity of demand measures how strongly quantity demanded responds to price. Find each percentage change by the midpoint method, divide the quantity figure by the price figure, and report the size of the result.

The first example uses the smallest clean numbers, so only the method is new.

Each line of arithmetic is shown, with its reason.

You should now be able to turn a price change and a quantity change into one number.

The notebook schedule gives a realistic example with less tidy numbers.

Recall: demand schedule, Chapter 4, section 4.1 'Quantity Demanded and the Law of Demand'.

The averages 35 and 45 come from the quantities and the prices.

Five slips account for most wrong answers.

Our first example uses clean numbers, so that the method is the only new thing. It is an everyday setting, not the running case. *Given:* a tea stall raises its price per cup from Tk 9 to Tk 11, and the quantity demanded falls from 24 cups to 16 cups an hour. *Find:* the price elasticity of demand. *Formula:* $E = \% \Delta Q \div \% \Delta P$, with midpoint percentages.

Steps. First, the quantity. The change is $16 - 24 = -8$ cups. The average is $(24 + 16) \div 2 = 20$ cups. So $\% \Delta Q = -8 \div 20 \times 100 = -40\%$. Second, the price. The change is $11 - 9 = \text{Tk } 2$. The average is $(9 + 11) \div 2 = \text{Tk } 10$. So $\% \Delta P = 2 \div 10 \times 100 = +20\%$. Third, divide: $E = -40 \div +20 = -2$, and the size is 2. *Answer:* the price elasticity of demand is 2, so quantity demanded changes twice as fast as price. *Check:* a 20% rise in price led to a 40% fall in quantity, and 40 is larger than 20, so a result above 1 is sensible.

PAUSE AND CHECK 5.P1 A good's price rises from Tk 8 to Tk 12, and the quantity demanded falls from 30 to 20. Find $\% \Delta P$, $\% \Delta Q$ and E .

You should now be able to say how a change in price and a change in quantity become one number: find each percentage change from the average of its old and new values, then divide quantity by price.

A REALISTIC EXAMPLE: THE NOTEBOOK

Case file, illustrative. Meghna and Sons, a bookshop in Dhanmondi, Dhaka, records a demand schedule for its popular 200-page notebook by the rule quantity = 80 - price, with price in taka. The price falls from Tk 50 to Tk 40, and the quantity demanded rises from 30 to 40. By the midpoint method, the percentage change in quantity is $10 / 35 = 28.57\%$, and the percentage change in price is $10 / 45 = 22.22\%$. The price elasticity of demand is 1.29.

Follow the arithmetic slowly. The two quantities are 30 and 40, so their average is 35, and the change of 10 gives $10 \div 35 = 28.57\%$. The two prices are 50 and 40, so their average is 45, and the change of 10 gives $10 \div 45 = 22.22\%$. Dividing, $28.57 \div 22.22 = 1.29$. *Check:* quantity moved by a larger percentage than price, so the result must lie above 1, and it does.

Common slips are few. Students divide price by quantity instead of quantity by price. They divide by the starting value in one place and by the average in another. They mix a figure in taka with a figure in units. They forget the units of time, comparing a weekly figure with a monthly one. And they report a negative sign in one answer and drop it in the next. Decide on the order and the sign rule before you begin.

The simplest way to record demand is a table. A **demand schedule** is a table showing the quantity demanded at each price. We now use the case of the shop. The next paragraph is a record, and its figures are fixed.

Case file, illustrative. Meghna and Sons records the weekly demand for its 200-page notebook. At a price of Tk 60 the quantity demanded is 20 notebooks; at Tk 50, 30; at Tk 40, 40; at Tk 30, 50; at Tk 20, 60. The quantity demanded equals 80 minus the price in taka.

WEEKLY DEMAND SCHEDULE FOR THE 200-PAGE NOTEBOOK

PRICE (TK)	QUANTITY DEMANDED (NOTEBOOKS A WEEK)
60	20
50	30
40	40
30	50
20	60

To read a row, take the line for Tk 40. It says that at a price of Tk 40, buyers choose to buy 40 notebooks a week. Read down the table and notice the pattern: as the price falls by Tk 10 each time, the quantity demanded rises by 10 each time. The rule in the case file, quantity equals 80 minus the price, gives the same numbers: 80 minus 40 is 40.

The pattern has a name. The **law of demand** says that when the price of a good rises, the quantity demanded falls, if other conditions stay the same. The final clause matters. Income, tastes and the prices of other goods must stay fixed, or we cannot tell whether a change in the quantity bought came from the price.¹ The law also works in the other direction: when the price falls, the quantity demanded rises.

Why should buyers behave in this way? Two reasons are enough for now. The first links to the last chapter.² Each extra notebook gives a buyer less satisfaction than the one before, so a buyer will take a further notebook only if its price is lower. The second reason is that a lower price brings in new buyers who could not afford the good before, and it tempts existing buyers to find new uses. A cheap notebook may become a diary, a ledger and a gift. **PAUSE AND CHECK 4.P1** At a price of Tk 50 the quantity demanded is 30 notebooks. Without drawing anything, say what the quantity demanded is at Tk 30, and state in one sentence the law of demand that your answer shows.

A table is exact, but a picture shows the pattern at a glance. A **demand curve** is a line on a graph showing the quantity demanded

¹ Recall: ceteris paribus, which means other things remaining equal, Chapter 1, 'What Economics Is', section 1.4 'Economic Laws'.

² Recall: diminishing utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.2 'The Law of Diminishing Marginal Utility'.

³ See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

Demand schedule. A table showing the quantity demanded at each price.

A demand schedule is a table of prices and the quantities demanded at them.

The case file records the weekly demand for the shop's notebook.

The same record is laid out as a table.

Each row pairs one price with one quantity demanded.

GO SLOWLY HERE.

Law of demand. When the price of a good rises, the quantity demanded falls, if other conditions stay the same.

The pattern in the table is the law of demand.

Two reasons explain why a lower price brings a larger quantity demanded.

Demand curve. A line on a graph showing the quantity demanded at each price.

YOU WILL NEED

Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand', for demand and its curve; Chapter 4, section 4.2 'Movement and Shift', for the difference between a movement along a curve and a shift; and Appendix A1, 'Maths You Will Use', for percentages and fractions.

Two goods lose a fifth of their price, yet they sell very differently.

The law of demand gives the direction of the change but not its size.

A price cut brings more customers but less money from each one.

Picture a street in Dhanmondi, Dhaka, on a Friday evening. One stall sells packets of salt. Beside it, a shop sells fashionable shoes. On the same day, both owners cut their prices by one-fifth. The salt seller waits all evening for a rush of customers, but it does not come. Families buy the same few packets they always buy, because a household cannot use more salt merely because salt is cheaper. The shoe seller finds a queue at the door by seven o'clock.

The law of demand, which you met in Chapter 4, says that a lower price raises the quantity demanded. It holds in both shops. Yet it says nothing about how much the quantity demanded rises. Salt rose by a trifle; shoes rose by a great deal. For an owner who must decide whether to cut a price, the direction is not enough. The owner needs a number that tells how strongly consumers respond.

There is a second puzzle behind the first. A lower price attracts more customers, but each customer pays less. Whether the shop finishes the day with more money received or with less depends on which effect is larger, and that depends on the same strength of response. This chapter builds the number. Section 5.1 defines and calculates it. Section 5.2 teaches you to read it. Section 5.3 explains what makes it large or small. Section 5.4 extends the idea to income and to the price of a rival good.

* * *

5.1 PRICE ELASTICITY OF DEMAND

WHY WE NEED PERCENTAGES

Begin with a household question. Suppose your monthly rent rises by Tk 500, and in the same month the price of a pen rises by Tk 5. Both rises look small, yet they are very different. Tk 500 added to a rent of Tk 20,000 is a rise of one-fortieth. Tk 5 added to a pen that costs Tk 10 is a rise of one-half. Only the change measured against the starting level shows how large the change is. For this reason economists measure changes in percentages.

Quantity has the same difficulty. Ten extra notebooks in a week is a large rise for a shop that sells thirty and a tiny rise for a shop that sells three thousand. A measure of response must therefore compare a percentage change in quantity with a percentage change in price. Such a measure has no units. It can be compared across goods, shops and countries.

GO SLOWLY HERE.

Money amounts and unit counts hide the real size of a change.

Response. Word in use. In daily life, an answer or a reply. In economics, the change in quantity demanded that follows a change in price or in another cause.

Quantity needs the same treatment: ten more units mean different things to different shops.

THE DETERMINANTS OF DEMAND, WITH A NOTEBOOK EXAMPLE

DETERMINANT	EXAMPLE FOR THE NOTEBOOK	EFFECT ON DEMAND
Buyers' incomes	Families have more money to spend	Rises
Prices of related goods	A rival notebook becomes cheaper	Falls
Tastes	Schools ask pupils to keep notebooks	Rises
Number of buyers	A new college opens nearby	Rises
Expectations	Buyers expect a price rise next month	Rises now

To read a row, take the second line: when a rival notebook becomes cheaper, some buyers leave, and demand for our notebook falls. The word *falls* here means that fewer notebooks are wanted at every price, not only at one price.

When a determinant changes, the whole curve moves to a new position. We call this a shift of the curve, which some books call a change in demand. This book uses the word shift. The case file gives a precise example.

Case file, illustrative. The price of a rival notebook falls by Tk 10. Weekly demand for the shop's notebook moves down by 5 at every price. At Tk 60 the quantity demanded becomes 15; at Tk 50, 25; at Tk 40, 35; at Tk 30, 45; at Tk 20, 55.

The new quantities are 5 smaller than the old ones in every row, so the curve moves to the left. A move to the left means a smaller quantity at every price. A move to the right means a larger quantity at every price. The picture shows the old curve and the new curve, and the arrow shows the direction.

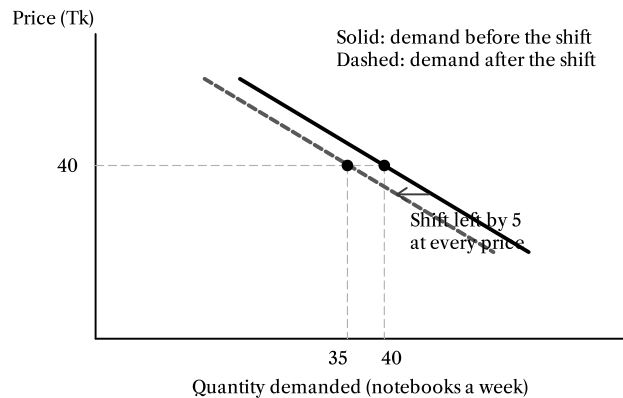


FIG. 4.2. The demand curve after a shift to the left, caused by a cheaper rival notebook

Read the picture along a horizontal line at Tk 40. The old curve meets this line at 40 notebooks. The new curve meets it at 35. The

Read each row as a change of condition and the resulting change of demand.

GO SLOWLY HERE.

A change in a determinant shifts the whole curve.

The case file records a shift caused by a rival's price.

The new curve lies to the left of the old curve.

Both curves slope downward, and the gap between them is 5 notebooks.

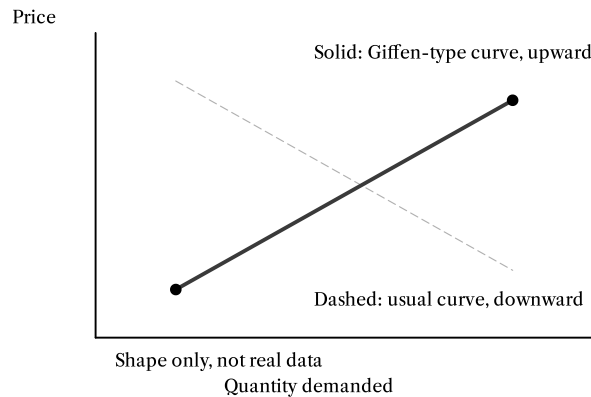


FIG. 4.3. *A Giffen-type curve slopes upward, in contrast with the usual curve*

The picture contrasts the usual curve, which slopes downward, with a Giffen-type curve, which slopes upward. On the upward line, a higher price goes with a larger quantity demanded. The numbers on this picture are not real data; it shows only the shape. Note one fact that keeps the law of demand valuable: the Giffen good is a rare, special case, and the law holds for almost every good that you will meet.

Two further cases depend on what other buyers do. The **snob effect** is a fall in demand for a good because many other people now buy it. A buyer who values owning something rare may leave a good once it becomes common. A scarf that a few people wore as a mark of taste loses its appeal when every street shows it.

The **bandwagon effect** is the opposite: a rise in demand for a good because many other people buy it. A mobile phone that all of a buyer's friends use becomes more attractive to that buyer, because it is easier to share messages and games. Both effects shift the curve, since they come from other buyers' choices and not from the good's own price.

Before moving on. Substitute goods replace each other, and complementary goods are used together. A normal good is bought more as incomes rise. A Giffen good is a rare exception to the law of demand, and the snob and bandwagon effects show that the choices of other buyers can shift demand. You should now be able to name one pair of each kind.

WORKED CASE: USING THE NOTEBOOK DEMAND SCHEDULE

The case file gives a rule: the quantity demanded equals 80 minus the price in taka. In symbols, $Q = 80 - P$, where Q is the quantity demanded in notebooks a week and P is the price in taka.

Rearranged $P = 80 - Q$ ⁴

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

The Giffen-type curve slopes upward, against the usual downward slope.

Snob effect. A fall in demand for a good because many other people now buy it.

The snob effect reduces demand when many others buy.

Bandwagon effect. A rise in demand for a good because many other people buy it.

The bandwagon effect raises demand when many others buy.

The section closes with the pairs and the exceptions.

Two links let us calculate any quantity or price from the demand rule.

The second example finds a price and then follows a shift.

Classification of a change as a movement or a shift is a matter of asking what changed.

Four common slips spoil these answers.

The chapter's main ideas in a few lines.

Two problems that apply the chapter to new settings.

A shift is often confused with a movement.

50) and 40 (at Tk 40) in the table, as it should for a price between Tk 50 and Tk 40.

Example 2. *Given:* the old rule $Q = 80 - P$, and the new rule after the rival's price fall, $Q = 75 - P$. *Find:* the price at which buyers choose 25 notebooks under each rule, and the change in the quantity demanded at Tk 40. *Formula:* $P = 80 - Q$ for the old rule, and $P = 75 - Q$ for the new rule. *Steps:* old rule, $P = 80 - 25 = \text{Tk } 55$; new rule, $P = 75 - 25 = \text{Tk } 50$; at Tk 40 the old quantity is $80 - 40 = 40$ and the new quantity is $75 - 40 = 35$, a change of $40 - 35 = 5$. *Answer:* 25 notebooks are chosen at Tk 55 before the shift and at Tk 50 after it, and at Tk 40 the quantity demanded is 5 notebooks lower. *Check:* the shift is 5 at every price, as the case file states.

Worked classification. *Situation:* three events occur in one month. The shop lowers its notebook price from Tk 50 to Tk 40. A rival lowers its price by Tk 10. A new college opens nearby. *Question:* which events are movements and which are shifts? *Reasoning:* in the first event, the price of the good itself changed. In the second, the price of a related good changed. In the third, the number of buyers changed. *Conclusion:* the first is a movement along the curve; the second and third are shifts. *Check:* only the first event changes the price on the vertical axis for our own notebook.

Common slips. Students often call a price change a shift. They mix the sides of the formula, using $Q = 80 - P$ when they need $P = 80 - Q$. They forget units, and they treat a fall in demand as a fall in the quantity demanded at only one price.

SUMMARY

Demand is the set of quantities buyers are willing and able to buy at each price, and a quantity demanded is one entry in that set. The law of demand says that, other conditions staying the same, a lower price brings a larger quantity demanded, so the demand curve slopes downward. A change in the price of the good is a movement along the curve. A change in a determinant is a shift of the curve. Substitute and complementary goods, normal goods, and the Giffen, snob and bandwagon cases show how demand depends on other goods, incomes and other buyers.

THINK IT THROUGH

4.T1 A jeweller finds that a certain luxury ring sells more when the price rises. Give two reasons why a buyer might behave in this way, and say whether the case breaks the law of demand.

4.T2 A bus fare rises, and at the same time the price of petrol falls. For bus travel, say which change is a movement along the curve and which is a shift, and give the direction of the shift.

WHERE PEOPLE GO WRONG

Error 1: "A fall in the price of the good shifts its curve." This is wrong.

and the horizontal axis shows the quantity bought. Each point on the line shows the highest price at which consumers would still buy that quantity. The dashed line is the market price. The consumers' surplus is the hatched triangle under the demand curve and above the price line. The dotted rectangle below it is the money that consumers actually pay.

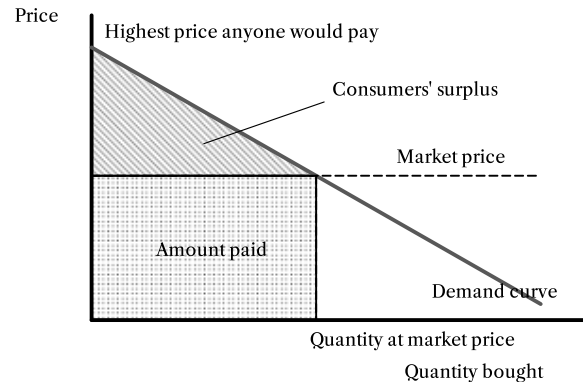


FIG. 6.1. *Consumers' surplus is the triangle under the demand curve and above the price*

When the demand curve is a straight line, the triangle has a simple area: half of the base multiplied by the height. The base is the quantity bought. The height is the highest price anyone would pay minus the market price. Suppose the highest price is Tk 100, the market price is Tk 60 and 40 units are bought. The surplus is $1/2 \times 40 \times (100 - 60) = 1/2 \times 40 \times 40 = \text{Tk } 800$.

Before moving on. Willingness to pay is the highest price a consumer would pay for one more unit. Consumers' surplus is the gap between that limit and the price actually paid, added over all units bought. On a chart it is the triangle under the demand curve and above the price line.

* * *

6.3 USING THE IDEA

Why measure a gain that nobody pays for? Because it shows the full value of a good to its users. A government that builds a bridge cannot judge it by the tolls alone, since the users gain more than they pay. A shopkeeper learns something too. Customers with a large surplus would have bought at a slightly higher price, so a price set too low gives away part of what the shop could earn. A price set too high loses the customers whose surplus was small.

The idea has limits, and the first concerns money. A taka means more to a poor consumer than to a rich one. Two consumers with the same surplus in taka may therefore gain different satisfaction. The measure shows the size of a gain in money. It does not show

ADVANCED NOTE: A BEGINNER MAY DEFER THIS PARAGRAPH.

For a straight demand curve, the surplus triangle has an area of half base times height.

Recap: willingness to pay, consumers' surplus and the triangle on the chart.

Surplus shows the full value of a good to its users, beyond the price they pay.

Surplus is measured in taka, and a taka means more to a poor consumer than to a rich one.

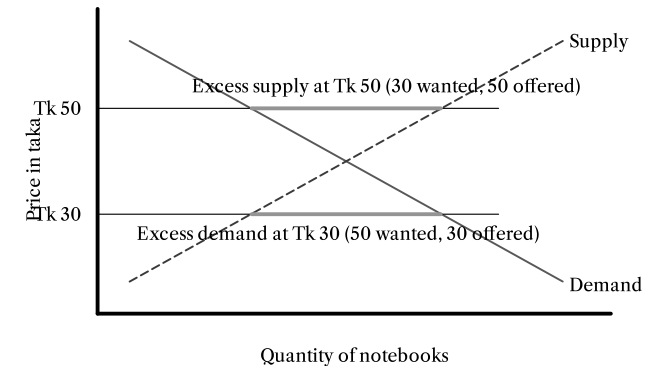


FIG. 9.2. *Shortage and excess supply at wrong prices*

Both forces push towards the same price, which is why the market settles there.

The rule has limits: adjustment takes time, and a controlled price blocks it.

Before moving on, recall the three ideas of this section.

The two forces point to the same place. Below the equilibrium price, excess demand pushes the price up. Above it, excess supply pushes the price down. Only at Tk 40 do the pushes stop. That is the answer to the opening puzzle: nobody decides the price, but every buyer and seller adds a small push, and the pushes balance at the equilibrium price.

The rule has limits. Adjustment takes time, and in a market for a perishable good the price may fall before the sellers understand why. A law that fixes the price blocks the pushes altogether; the Think It Through section returns to this. The rule also assumes a competitive market. Finally, from the owner's view the price is a signal, and from the consumer's view it is a limit on what can be bought.

Before moving on. A competitive market has many buyers and sellers, none of whom set the price. The equilibrium price makes the quantity demanded equal the quantity supplied. Above it there is excess supply and the price falls; below it there is excess demand and the price rises.

* * *

9.3 CHANGES IN EQUILIBRIUM

A school opens beside the shop, and a hundred more students need notebooks. The price of notebooks was Tk 40 yesterday. Will it still be Tk 40 today? The equilibrium price holds only while the conditions behind demand and supply stay the same. When a condition changes, a curve moves, and the market finds a new crossing point.

Two kinds of change must be kept apart. A *shift* means the whole demand curve or the whole supply curve moves, because something other than the price has changed, such as incomes or the price of materials. A *movement along* a curve happens when the price changes and the curve stays still. Section 9.2 showed move-

A puzzle: a school opens nearby, and the equilibrium price may no longer hold.

GO SLOWLY HERE.

A shift moves a whole curve; a change of price moves us along a curve.

Formula. The same as before: $S = (W1 - P) + (W2 - P) + \dots$ for every pen bought. *Steps at Tk 12.* The consumer willing to pay Tk 9 does not buy. $25 - 12 = \text{Tk } 13$. $18 - 12 = \text{Tk } 6$. $14 - 12 = \text{Tk } 2$. Sum: $13 + 6 + 2 = \text{Tk } 21$. *Steps at Tk 9.* All four consumers buy. $25 - 9 = \text{Tk } 16$. $18 - 9 = \text{Tk } 9$. $14 - 9 = \text{Tk } 5$. $9 - 9 = \text{Tk } 0$. Sum: $16 + 9 + 5 + 0 = \text{Tk } 30$. Rise: $30 - 21 = \text{Tk } 9$.

Answer. The surplus is Tk 21 at the price of Tk 12 and Tk 30 at the price of Tk 9, a rise of Tk 9. *Check.* The price fell by Tk 3, so each of the three old buyers gains Tk 3, and $3 \times 3 = \text{Tk } 9$. The new buyer adds Tk 0. The total agrees with the rise.

Three slips are common. The first is a wrong sign. Subtract the price from the willingness to pay, not the reverse, or the answer is negative. The second is to include a consumer who does not buy. Omit such a consumer, because the surplus from that unit is zero. The third is to forget the unit. Write Tk beside each figure, and do not mix prices from different dates.

* * *

SUMMARY

Consumption is the use of goods and services to satisfy wants. Consumers have limited income and spend it where an extra taka adds the most utility. Willingness to pay is the highest price a consumer would pay for one more unit. Consumers' surplus is the gap between willingness to pay and the price actually paid, and on a chart it is the triangle under the demand curve and above the price. A fall in price is a movement along the demand curve and it raises the surplus. In the notebook case, the surplus rose from Tk 30 to Tk 60 when the price fell from Tk 40 to Tk 30.

THINK IT THROUGH

6.T1 In the notebook case the surplus rose from Tk 30 to Tk 60 when the price fell from Tk 40 to Tk 30. Explain in your own words why the surplus rises when the price falls.

6.T2 A public bridge charges a toll of Tk 20, and 1,000 people cross each day. Many of them would have paid more. Explain why the tolls collected do not measure the benefit of the bridge to its users.

WHERE PEOPLE GO WRONG

A common belief is that the price shows what a good is worth to the consumer. It does not. The price shows only what the consumer must pay. The worth to the consumer is the willingness to pay, which is usually higher. The difference between the two is the surplus.

A second belief is that surplus is money that the consumer receives. No money changes hands. Surplus is satisfaction expressed in taka, and it stays in the mind of the consumer.

A third mistake is to include a consumer who does not buy. A

The same formula is applied twice, once at each price.

The surplus is Tk 21 at Tk 12 and Tk 30 at Tk 9, and the rise passes the check.

Three slips are common: wrong sign, buyers who do not buy, and missing units.

The main ideas of the chapter in a few lines.

A common belief is that the price shows what a good is worth to the consumer; it does not.

A second belief is that surplus is money received; it is satisfaction counted in taka.

A third mistake is to count a consumer who does not buy

Time is the strongest factor, followed by spare capacity and stocks.

The manager and the owner see the same factors from different positions.

Before moving on, recall the measure and the five types.

A puzzle: neither side can force the price on the other.

Market. Word in use. In daily life, a place where people meet to buy and sell, such as a street market. In economics, all the buyers and sellers of one good, wherever they are.

Competitive market. A market with many buyers and sellers, none of whom can set the price.

We begin with markets where nobody controls the price.

Equilibrium. Word in use. In daily life, a state of balance, such as a scale holding still. In economics, a price and quantity at which nothing pushes the price to change.

Equilibrium price. The price at which the quantity buyers want equals the quantity sellers offer.

The price where both sides agree is the equilibrium price.

The case file finds the equilibrium by setting the two quantities equal.

FACTORS

Four factors decide where a good lies in the table. *Time* matters most. In a day, a firm can only sell what it holds; in a year, it can build more machines. *Spare capacity* matters next: a printing firm with idle presses can raise output quickly. *Stocks* of finished goods allow a quick response, but perishable goods cannot be stored. *Inputs* matter too: if workers and materials are hard to find, output rises slowly.

A manager asks, "Can my workers and machines make more this week?" An owner asks, "Is it worth buying a new machine?" The first question concerns the short run and gives a low elasticity. The second concerns the long run and gives a higher one. Elasticity of supply is therefore never a fixed property of a good; it depends on the period we choose.

Before moving on. Elasticity of supply measures how strongly firms respond to a price change, using percentages and the midpoint. A value above 1 is elastic, 1 is unit elastic, and below 1 is inelastic. Time, spare capacity, stocks and inputs decide the value.

* * *

9.2 MARKET EQUILIBRIUM

Return to the market row. A seller wishes to charge Tk 60 for a notebook, but buyers will pay only Tk 30. Neither side can force the other. The price that finally appears comes from the meeting of both sides, and from neither side alone. To study that meeting, we must first say what kind of market we mean.

We study a **competitive market**. A competitive market means a market with many buyers and sellers, none of whom can set the price. A street of tea stalls selling the same tea is an example: if one stall raises its price, buyers walk to the next stall. A market where one seller does control the price is a different case, taught in Chapter 14, 'Monopoly and Imperfect Competition', section 14.1 'Monopoly'.

The price at which the market settles has a name: the **equilibrium price**. The equilibrium price means the price at which the quantity buyers want equals the quantity sellers offer. If buyers want 40 notebooks and sellers offer 40 notebooks, both sides are satisfied and nobody has a reason to change the price. The quantity bought and sold at that price is the *equilibrium quantity*.

Case file, illustrative. In the notebook market, the quantity demanded equals 80 minus the price, and the quantity supplied



CHAPTER VII

Indifference Curve Analysis

Can a shopper's wishes be drawn on paper?



This chapter draws a consumer's taste as a line of equal satisfaction and her purse as a line of limited spending. Where the two lines meet in the right way, we find the combination of two goods that suits her best.

YOU WILL NEED

Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; Chapter 8, 'Supply and the Supply Curve', section 8.1 'Quantity Supplied and the Law of Supply'; Chapter 5, 'Elasticity of Demand', section 5.1 'Price Elasticity of Demand'; and Appendix A1, 'Maths You Will Use'.

A seller must choose a price, yet no single seller truly decides it.

By evening the sellers charge almost the same price, and nobody announced it.

The chapter has two jobs: measure how sellers respond, then find where the market settles.

The direction of the response is known from Chapter 8; its size is not.

Elasticity of supply. The percentage change in quantity supplied divided by the percentage change in price.

A single number records how strongly sellers respond to a price change.

Recall: price elasticity of demand, Chapter 5, section 5.1 'Price Elasticity of Demand'.

Percentages let us compare goods that are measured in

Early on a market morning, a seller unloads crates of notebooks and must choose a price. If the price is too high, shoppers walk past and the crates stay full. If the price is too low, the crates are empty by ten o'clock, and she wonders how much money she gave away. Every seller in the row faces the same worry, and each one watches the others.

By evening, a strange thing has happened. Nobody called a meeting and nobody gave an order, yet the sellers in the row charge almost the same price, and most of the crates are sold. Something has guided hundreds of separate choices to one number. What is that something? Why did the price settle at this number and not at another?

This chapter answers in two steps. First, we measure how strongly sellers react when the price changes; section 9.1 gives this measure its name. Second, we join the buyers' side and the sellers' side to find the price where the market settles, and we ask what moves that price. Many students find the second step easier once the first is secure, because both steps use the same idea of percentage change.

* * *

9.1 ELASTICITY OF SUPPLY

Chapter 8 taught the law of supply: when the price of a good rises, firms offer more of it. That law gives a direction and nothing more. Suppose the price of a notebook rises by 10%. Will firms offer 2% more, or 20% more? A baker with a hot oven can bake extra loaves by the afternoon. A farmer who planted onions in March cannot grow more by Friday. Both respond in the same direction, yet the size of the response differs greatly. To compare them, we need one number that records the size.

That number is called **elasticity of supply**. Elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. Suppose the price rises by 10% and the quantity supplied rises by 5%. Then the elasticity of supply is $5 \div 10 = 0.5$. Because supply slopes upward, price and quantity supplied move in the same direction, so the answer is never negative.

Why do we use percentages and not plain changes? A change of 10 notebooks is large for a small shop and tiny for a printing firm. A change of 10 kilograms means little for rice and much for saffron. A percentage compares the size of the move with the starting amount. The measure copies the one you met for buyers in

combination on one curve has the same utility, whatever that utility is worth in numbers.

Many students find the word *curve* confusing, because it suggests a bend, and the economist's curve is sometimes straight. Treat it as a name for any line drawn on a graph to show a relationship. An indifference curve does bend, for a reason that we meet in section 7.2. Until then, think of it only as a path through the combinations that please equally.

Now build one indifference curve from small, round numbers. The table below lists five combinations of notebooks and pens that one student values equally. Read each row from left to right: the letter names the combination, and the next two columns give its notebooks and its pens. These figures are made up to teach the idea; they are not part of the case file. Notice how the number of pens falls as the number of notebooks rises.

ONE STUDENT'S EQUALLY SATISFYING COMBINATIONS (ILLUSTRATION)

COMBINATION	NOTEBOOKS	PENS
A	3	20
B	4	14
C	6	8
D	8	5
E	12	2

A graph shows the same facts in a second form. The first time we use one, here is how to read it. The horizontal axis counts notebooks. The vertical axis counts pens. Each point is one combination: its distance along tells the notebooks, and its height tells the pens. Join the five points with a smooth line and you have an indifference curve. It falls from left to right and bends in towards the corner where the axes meet.

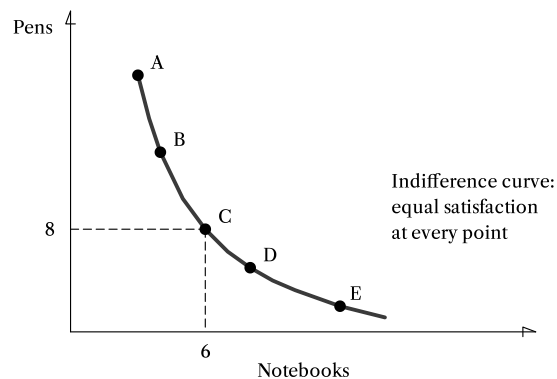


FIG. 7.1. An indifference curve through the five combinations A to E

Why does the curve fall? Suppose the student holds combination A and receives one more notebook. She now has more of one good, so she is better off. To return to her old level of satisfaction she

Curve. Word in use. In daily life, a bend in a road or a river. In economics, a line drawn on a graph, which may bend or may be straight.

The word curve means a line on a graph, bent or straight.

A table of five combinations shows the same satisfaction throughout.

The same five combinations drawn as points make a falling line.

The curve falls because more of one good must replace less of the other.

pens are few and notebooks many, each pen is precious, so she gives up only a few. This falling rate is why an indifference curve bends in towards the corner of the axes: the curve grows flatter as we move to the right.

So far we measured the rate over a step. At a single combination, the economist measures it for a very small step, and the result equals the steepness of the curve at that point. At combination C the steepness is 2, which lies between the rate 3 on its left and the rate 1.5 on its right. This is the value that appears in the case of section 7.4. Common slips are easy to make here: dividing in the wrong order, putting the goods on the wrong axes, and forgetting to say "pens per notebook".

PAUSE AND CHECK 7.P2 Along an indifference curve the student holds more notebooks and fewer pens. Why does she give up fewer pens for each extra notebook?

Before moving on. The marginal rate of substitution is the number of pens a consumer gives up for one more notebook, with satisfaction unchanged. It falls along a curve, which is why the curve bends inwards. *You should now be able to say* how to compute the rate between two combinations on one curve.

7.3 THE BUDGET LINE

Taste is only half the story. Return to the stall with Tk 100. Suppose mangoes cost Tk 20 each and guavas cost Tk 10 each. Spending all the money on mangoes buys 5. Spending it all on guavas buys 10. Three mangoes and 4 guavas cost Tk 60 and Tk 40, which is Tk 100 again. Each of these combinations is affordable, and together they follow a pattern that we can draw.

A **budget line** is a line showing every combination of two goods that a consumer can buy with a given income at given prices. At the stall, the points for 5 mangoes, 10 guavas, and 3 mangoes with 4 guavas all lie on one budget line. Points inside the line are affordable but leave some money unspent. Points beyond the line cost more than the shopper has. The word *line* here means a path of points on a graph, and this one is straight.

Case file, illustrative. A student has Tk 400 to spend on notebooks at Tk 40 each and pens at Tk 20 each. The most that can be bought is 10 notebooks or 20 pens.

Given. Income Tk 400; notebook price Tk 40; pen price Tk 20. *Find.* The most of each good, and the pens given up for each extra notebook. *Formula.* In words: the most of a good is income divided by its price; the pens given up for one notebook are the notebook price divided by the pen price. In symbols: $M = I \div P$, where M is the most of a good, I is income and P is that good's price; price ratio = notebook price $\div$ pen price. *Steps.* Notebooks: $\text{Tk } 400 \div \text{Tk } 40 = 10$. Pens: $\text{Tk } 400 \div \text{Tk } 20 = 20$. Price ratio: $\text{Tk } 40 \div$

At one combination, the rate is the steepness of the curve there.

Recap of section 7.2.

A fixed sum of money limits what any shopper can buy.

Budget line. A line showing every combination of two goods that a consumer can buy with a given income at given prices.

Line. Word in use. In daily life, a queue of people or a mark made with a pen. On a graph, a path of points; the budget line is straight, while an indifference curve bends.

A budget line shows every combination that the money can buy.

The case file gives the student's income and the two prices.

A worked example finds both ends of the line and its steepness.

SUMMARY

Supply is the set of quantities that sellers are willing and able to sell at each price, and a quantity supplied is one entry in that set. The law of supply says that, other conditions staying the same, a higher price brings a larger quantity supplied, so the supply curve slopes upward. A change in the price of the good is a movement along the curve. A change in the cost of inputs, technology, taxes or subsidies, or the number of sellers is a shift of the curve. An exceptional supply curve, such as a backward-bending curve or a fixed supply, does not slope upward.

THINK IT THROUGH

8.T1 A worker is offered more and more pay for each hour, yet at a very high rate chooses to work fewer hours. Give two reasons for this choice, and say what shape of supply curve it makes.

8.T2 A fishmonger has 200 fish on a market day and no way of getting more before the market closes. Describe the supply curve for that day, and say what changes if the market is open for a week.

WHERE PEOPLE GO WRONG

Error 1: "A rise in the price of the good shifts its supply curve." This is wrong. A rise in the price of the good moves the point along the same curve. Only a change in a determinant shifts the curve.

Error 2: "A tax on inputs shifts supply to the right." This is wrong. A tax raises the seller's expense, so fewer units are offered at every price, and the curve shifts to the left.

Error 3: "Supply and quantity supplied mean the same." This is wrong. Quantity supplied is one number at one price. Supply alone is the whole list of prices and quantities.

Error 4: "The cost of a notebook is what the shop pays the wholesaler, so cost and price are the same." This is wrong in this book. Price is what a buyer pays. Cost is the seller's expense in making and delivering the good.

IN HISTORY

In 1890, Alfred Marshall, who lived from 1842 to 1924 and taught economics at Cambridge in England, published *Principles of Economics*. He had begun the book in 1881. Before him, economists had long argued about what governs the price of a good: the satisfaction it gives to buyers, or the expense of making it. Marshall answered with a comparison. We might as sensibly ask which blade of a pair of scissors cuts a piece of paper, he wrote, as ask which of the two decides the price.

Marshall's point was that supply and demand work together, and both are needed. The book became the leading textbook in England for many years. To this day, an economist who wishes to explain why the price of a good has changed begins by asking whether the demand curve shifted or the supply curve shifted.

The chapter's main ideas in a few lines.

Two problems that apply the chapter to new settings.

A shift is often confused with a movement.

The direction of a shift is often reversed.

Supply is not the same as quantity supplied.

Cost and price are different ideas in this book.

An English professor compared supply and demand to the two blades of a pair of scissors.

The picture has lasted, and modern economists still begin with it.

The steepness of the curve is the marginal rate of substitution, and the steepness of the line is the price ratio. At the touching point the two are equal. The rule has a plain meaning. The curve's steepness says how many pens she would give up for one more notebook. The line's steepness says how many pens she must give up. If the first is larger, a notebook is worth more to her than it costs, so she buys more notebooks; her willingness to pay, measured in pens, exceeds the price.³ If the first is smaller, she buys fewer. She stops only when the two are equal.

IN SHORT

At consumer equilibrium the marginal rate of substitution equals the price ratio, and the combination lies on the budget line.

The rule needs care. If the student would rather spend everything on one good, the best point lies on an axis, and the two steepnesses need not be equal. The rule also assumes that all income is spent and that the curves bend in as described in section 7.1. This chapter treats only the usual case, in which both goods are bought.

Before moving on. Consumer equilibrium is the point on the budget line that lies on the highest reachable curve. There the curve touches the line, and the marginal rate of substitution equals the price ratio. *You should now be able to say* why a combination on a crossing curve is not the best.

A WORKED CASE: THE STUDENT'S BEST COMBINATION

Case file, illustrative. A student has Tk 400 for notebooks at Tk 40 and pens at Tk 20. The most that can be bought is 10 notebooks or 20 pens. The best bundle is 6 notebooks and 8 pens (Tk 240 + Tk 160 = Tk 400), where the marginal rate of substitution is 2, equal to the price ratio $40 / 20 = 2$.

Given. Income Tk 400; notebooks Tk 40; pens Tk 20; the curve table of section 7.1; the claimed best combination of 6 notebooks and 8 pens. *Find.* Whether this combination is the best the student can afford. *Formula.* In words: spending equals notebooks times their price plus pens times their price; the best combination has its rate equal to the price ratio. In symbols: $S = (N \times \text{Tk } 40) + (K \times \text{Tk } 20)$, where S is spending, N is notebooks and K is pens.

Steps. (1) Spending: $S = (6 \times 40) + (8 \times 20) = 240 + 160 = \text{Tk } 400$, so the combination lies on the budget line. (2) Price ratio: $40 \div 20 = 2$. (3) The rate at 6 notebooks and 8 pens is 2, as section 7.2 found. (4) The rate equals the price ratio, so the curve touches the line. (5) Test neighbours on the line. P is 4 notebooks and 12 pens: $160 + 240 = \text{Tk } 400$. The table says 4 notebooks with 14 pens is equally good as the claimed combination, and P has 2 pens fewer, so P is worse. Q is 8 notebooks and 4 pens: $320 + 80 = \text{Tk } 400$. The table says 8 notebooks with 5 pens is equally good, and Q has 1 pen fewer so Q is worse. *Answer.* Yes, 6 notebooks and 8 pens is the best combination. ³ Recall: willingness to pay and consumers' surplus, Chapter 6, 'Consumption and Consumers' Surplus', section 6.2 'Consumers' Surplus'.

At the best point, the rate she accepts equals the rate the market asks.

The rule has exceptions, so check its conditions.

Recap of section 7.4.

The case file states the student's best combination.

We test the case file's claim step by step.

The steps check spending, steepness and the two nearest affordable neighbours.

is 10 notebooks at every price. The price on the vertical axis did not change, so the shift is not a movement along either curve. A move to the left means less supplied at every price. A move to the right means more supplied at every price.

PAUSE AND CHECK 8.P2 The government gives a subsidy to the makers of paper. Is the result a movement along the supply curve for notebooks or a shift? Say in which direction it goes, and give one reason.

Before moving on. Ask one question: what changed? If the price of the good itself changed, there is a movement along the supply curve. If the cost of inputs, technology, a tax or subsidy, or the number of sellers changed, there is a shift of the whole curve. A fall in expenses or a rise in the number of sellers shifts the curve to the right, and the opposite changes shift it to the left.

8.3 EXCEPTIONAL SUPPLY CURVES

The law of supply describes most goods, but not all. Some supply curves do not slope upward, and economists have a name for them.

An **exceptional supply curve** is a supply curve that does not slope upward, such as one that bends backward or stays fixed. Two cases are worth knowing. Each is a limit of the law, and each has a reason that you can state in one sentence.

The first case is the backward-bending curve. The best example is the supply of working hours by one person. As the pay for each hour rises, the worker at first offers more hours, because work is now better rewarded. At a very high rate of pay, however, the worker can earn the income that he or she needs in fewer hours, and may prefer more rest. The hours offered then fall as the pay rises further. The curve rises, reaches a peak and then bends back.

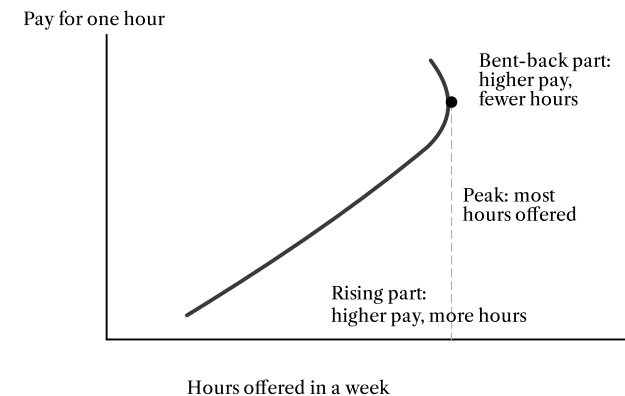


FIG. 8.2. A backward-bending supply curve of working hours

Read the chart in the same way as before. The vertical axis shows

A short test separates the two causes.

Not every supply curve rises, and the exceptions teach the reasons behind the rule.

Exceptional supply curve. A supply curve that does not slope upward, such as one that bends backward or stays fixed.

An exceptional supply curve does not rise.

A backward-bending curve appears when a higher price makes a seller supply less.

The curve rises at first and

IN HISTORY

The curves in this chapter are older than the 1930s. Francis Edgeworth, an economist who taught at Oxford, drew the first indifference curves in his book *Mathematical Psychics* in 1881. The Italian economist Vilfredo Pareto used them again in his *Manuale di economia politica* in the early 1900s, and he stressed that a theory of choice needs only the comparison of combinations.

In 1934 two economists at the London School of Economics, John Hicks and R. G. D. Allen, published *A Reconsideration of the Theory of Value*. It appeared in two parts in the journal *Economica*, in February and May 1934. They credited Pareto as their inspiration and rebuilt the theory of consumer choice on indifference curves, so that the argument no longer needed satisfaction to be counted in numbers. The budget line, the marginal rate of substitution and the tangency rule of this chapter are the core of that theory, and the method is still the usual way to teach consumer choice.

Indifference curves were drawn before 1934.

In 1934 Hicks and Allen rebuilt consumer theory on the curves.

REVIEW QUESTIONS

- 7.1 A student has Tk 200. Notebooks cost Tk 40 and pens cost Tk 20. Find the most of each good she can buy, and name two other combinations on her budget line.
- 7.2 From combination C (6 notebooks, 8 pens) the student moves to combination D (8 notebooks, 5 pens) and stays equally satisfied. Find the marginal rate of substitution over this step. Is it larger or smaller than the rate of 3 from B to C, and why?
- 7.3 Income is Tk 450, notebooks cost Tk 50 and pens cost Tk 30. Find where the budget line meets each axis, and its steepness.
- 7.4 Income is Tk 400 and pens cost Tk 20. The price of a notebook rises from Tk 40 to Tk 50. Find the new ends of the budget line and say how the line moves.
- 7.5 Notebooks cost Tk 40 and pens Tk 20. At 4 notebooks and 12 pens, which lie on her budget line, her marginal rate of substitution is 4. Is she at consumer equilibrium? If not, which good should she buy more of?
- 7.6 At 6 notebooks and 8 pens her marginal rate of substitution is 2 pens per notebook, and a pen costs Tk 20. Find in taka the most she would pay for one more notebook. Compare it with the price of Tk 40 and say what this shows about consumers' surplus on that notebook.

Supply curve. A line on a graph showing the quantity supplied at each price.

A supply curve draws the table as a line on a graph.

The supply curve rises from the bottom left to the top right.

The section closes with the four ideas in a few lines.

A tax on paper changes what the wholesaler offers, although the notebook's price has not moved.

A change in the price of the good itself moves us along one curve.

Four conditions decide where the whole supply curve lies.

Dearer inputs raise the seller's expense and reduce what is offered.

A table is exact, but a picture shows the pattern at a glance. A **supply curve** is a line on a graph showing the quantity supplied at each price. It is drawn in the same way as the demand curve.² Price is on the vertical axis, and the quantity supplied is on the horizontal axis. Each row of the table gives one point: the price in the row sets the height, and the quantity sets the distance from the vertical axis. Join the five points, and a line appears.

For the wholesaler, the five points lie on a straight line that rises from the bottom left to the top right. We call it a curve although it is straight, because the word covers lines of every shape. The upward slope is the law of supply drawn as a picture: a higher price goes with a larger quantity supplied. A demand curve slopes downward, and a supply curve slopes upward. That single difference marks the two sides of every market, and the chart in the next section shows both the curve and what moves it.

Before moving on. Quantity supplied is one number at one price. Supply is the whole list of prices and quantities, which the supply schedule records and the supply curve draws. The law of supply says that, other conditions staying the same, a higher price brings a larger quantity supplied.

8.2 DETERMINANTS OF SUPPLY

Suppose the government places a tax on paper, the main material in a notebook. The wholesaler's expenses rise at once. The price of the notebook has not changed, and yet the wholesaler now offers fewer notebooks at the same price. This event is different from a change in the price of the notebook, and it shows how supply has its own movements and shifts.³

First, the price of the good itself changes. If the notebook's price rises from Tk 40 to Tk 50, the quantity supplied rises from 40 to 50 notebooks a week. On the graph, the point moves along the same line. We call this a movement along the curve. The curve does not change position, because the whole list of prices and quantities is the same list as before. This is the mirror of the movement along a demand curve.

Second, something other than the price changes. The supply curve was drawn with other conditions held fixed, and four are most important. The first is the cost of inputs: the materials, labour, rent and power that a seller must buy. The second is technology, meaning the methods and machines used in making the good. The third is taxes and subsidies. The fourth is the number of sellers in the market. When any of these changes, sellers offer different quantities at every price.

Cost of inputs. If paper, wages or rent become dearer, each notebook is more expensive to make, and the wholesaler offers fewer at every price. If inputs become cheaper, the wholesaler offers more. The effect works through the seller's expense, which is why

² See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

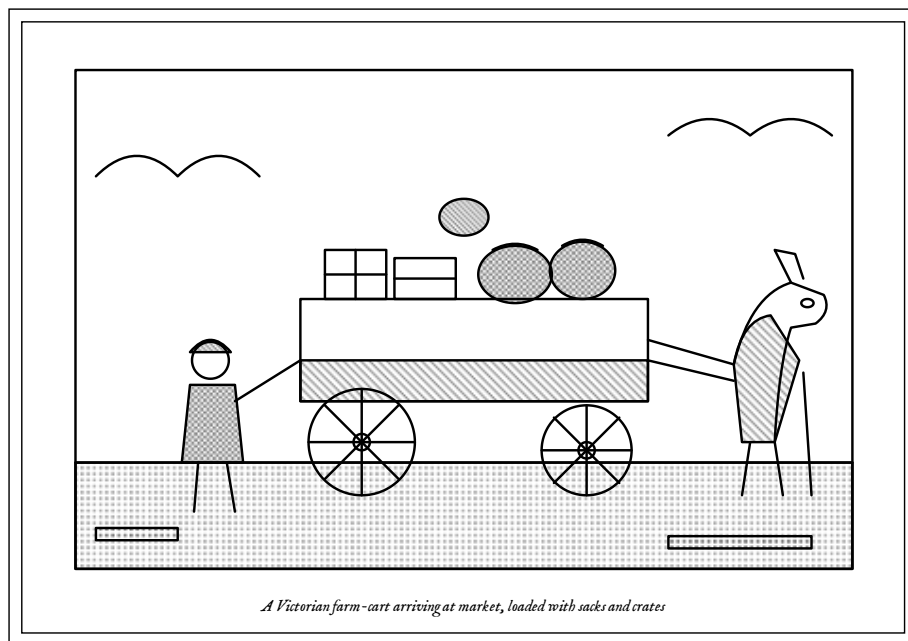
³ Recall: shifts, Chapter 4, 'Demand and the Demand Curve', section 4.2 'Movement and Shift'.

Two ideas from Chapter 4 and one appendix help with this chapter.

A farm-cart arrives at the market town, and its load depends on the price.

The same farmer brings different loads, and he cannot bring an unlimited load.

Chapter 4 told the story of buyers; this chapter tells the story of sellers.



CHAPTER VIII

Supply and the Supply Curve

Why does a farmer bring more sacks to market when the price is high?



This chapter builds the seller's side of the market. It explains what sellers are willing and able to sell at each price, why a higher price brings a larger quantity, how to draw the supply curve, and what shifts it. It ends with a few exceptional curves that do not rise.

You will need two ideas from Chapter 4: what buyers are willing and able to buy, from Chapter 4 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; and the difference between a movement along a curve and a shift, from Chapter 4 'Demand and the Demand Curve', section 4.2 'Movement and Shift'. Appendix A1 'Maths You Will Use' helps with reading a graph and with rearranging an equation.

Picture a farm-cart arriving at a Victorian market town on a Saturday morning. It is loaded with sacks of potatoes and crates of eggs. The farmer has driven eleven miles in the dark, and he has chosen carefully what to bring. Last month, when potatoes sold poorly, he brought a few sacks and kept the rest to feed his pigs. This month the price is high, and the cart is piled so high that the horse leans into the harness.

The farmer, the farm and the horse are the same as last month. Only the price has changed. Yet he brings more sacks, and the market sees a larger load. Notice also what he does not do. Even at a very high price he does not bring a hundred sacks, because he has only so many fields, so many workers and so many hours. Why does a higher price bring a larger load, and what limits it?

Chapter 4 told the story of the crowd that buys. This chapter tells the story of the farmer who sells. The two stories mirror each other, and many ideas return in reverse form. That mirror makes the chapter easier, but it also makes a trap: students sometimes copy a rule from demand and forget to reverse it. We therefore build the seller's side step by step, and we mark each place where it differs from the buyer's side.

* * *

8.1 QUANTITY SUPPLIED AND THE LAW OF SUPPLY

Begin with a question about a seller: how many would you offer at each price?

Supply. The quantities of a good that sellers are willing and able to sell at each price in a period.

Supply is the whole list of prices and quantities supplied, not one number.

Supply. Word in use. In daily life, supply can mean a store

Begin with a question from your own life. Suppose you could make bracelets at home and sell them to friends. At a price of Tk 20 each you might make a few, because the beads and the time are worth more than the price. At Tk 60 each you would make many more. The number of units that sellers choose to offer for sale at one particular price in a period is called the quantity supplied. It is one number tied to one price, in the same way as the quantity demanded.

Economists use the word **supply** for the larger idea. Supply means the quantities of a good that sellers are willing and able to sell at each price in a period. As with demand, two conditions are needed. A seller who wishes to sell but has no stock and no means of making more is not supplying anything. A seller with plenty of stock who refuses to sell is not supplying either. For the notebook at the shop, supply is the full list: so many notebooks at Tk 20, so many at Tk 30, and so on. One entry is a quantity supplied. The whole list is supply.

The everyday word needs care. In daily speech people say that "supply is short" when they mean that the stock is small today. This

corrected rule: price changes cause movements along; other changes cause shifts.

Mistake: "A flat supply curve is always elastic." Steepness depends on the units of the axes, but elasticity uses percentages and does not. A straight line from the origin has an elasticity of 1 at every point, however steep it looks.

Mistake: "At equilibrium everyone is satisfied." Equilibrium means only that no force pushes the price to change. Some buyers still cannot afford the good at that price, and some firms would like a higher price. The corrected rule: equilibrium describes balance, not fairness.

IN HISTORY

On 29 September 2019, the government of India banned the export of onions with immediate effect, to improve supply at home. Bangladesh imports about a quarter of the onions it uses, and the Commerce Minister of Bangladesh said that about nine-tenths of those imports came from India. The ban therefore shifted the supply of onions in Dhaka to the left. In the Kawran Bazar market, the price of imported onions rose from Tk 60 to Tk 100 a kilogram overnight, according to the *Business Standard*. In the following weeks, retail prices in Dhaka reached about Tk 250 a kilogram, according to the *Dhaka Tribune*.²

The case also shows the factors of section 9.1. In the short run, the supply of onions was inelastic: the harvest season had ended and the crop could not be grown again in a week. Buyers kept buying, because onions are a daily kitchen good. Over the following weeks, importers brought onions from other countries, including Pakistan and Egypt, and supply became more elastic. As the new onions arrived, the price eased.

REVIEW QUESTIONS

9.1 The price of a good rises from Tk 20 to Tk 30, and the quantity supplied rises from 90 to 110 units. Calculate the elasticity of supply with the midpoint method, and name its type.

9.2 The price of mangoes rises from Tk 100 to Tk 130 a kilogram. The weekly quantity supplied rises from 400 to 500 kilograms. Calculate the elasticity of supply.

9.3 The price of a good falls from Tk 60 to Tk 50, and the weekly quantity supplied falls from 330 to 270 units. Calculate the elasticity of supply, and name its type.

9.4 In a market, quantity demanded = 60 – price and quantity supplied = 2 × price. Find the equilibrium price and quantity. Then demand shifts to a quantity demanded of 90 – price; find the new equilibrium.

9.5 In the market of question 9.4, supply then shifts to a quantity supplied of 2 × price – 30, with demand at 90 – price. Find the new

² Prices differed by market, by variety and by day, so reports give different peaks.

Mistake two: reading elasticity as the steepness of the line.

Mistake three: thinking equilibrium means a fair price or a happy market.

In 2019, a sudden ban shifted supply and the price of onions in Dhaka jumped.

The case shows that short-run supply is inelastic and that time raises the response.

Five questions follow the ladder from clean figures to a combined problem.

* * *

SUMMARY

An isoquant shows all mixes of two inputs that give the same output. An isocost line shows all mixes that cost the same total amount, and its slope is the price ratio. The marginal rate of technical substitution is the amount of one input that can be saved when another is added, keeping output the same. The least-cost combination is where the lowest isocost line touches the isoquant, so the marginal rate of technical substitution equals the price ratio. In the case, the least cost for 100 books is Tk 4,000, at 4 workers and 4 machines. A wage rise moves the best mix toward machines.

THINK IT THROUGH

12.T1 The wage rises from Tk 600 to Tk 900 and the machine rent stays at Tk 400. Without calculating, predict how the least-cost mix changes. Then explain why total cost for 100 books rises by less than 4 × Tk 300.

12.T2 A firm has signed a contract for 6 machines and cannot change it this month. Can it reach the least-cost combination for 100 books in the case? Explain.

WHERE PEOPLE GO WRONG

A common belief is that a firm should use more of the input with the lower price. Price alone does not decide. The rule compares the output added per taka, so a dear input that adds a great deal of output can still be the better choice.

A second belief is that least cost means the fewest inputs. It does not. The mix (3, 6) uses nine inputs in all and costs Tk 4,200, while the mix (4, 4) uses eight and costs Tk 4,000, so counting inputs is a poor guide. Only the total bill, found from the prices, decides.

A third belief is that the best mix spends equal amounts on workers and machines. At (4, 4), the wage bill is Tk 2,400 and the rent bill is Tk 1,600. The rule requires equal output per taka, not equal spending.

A fourth mistake is to treat an isoquant as if it were a consumer's indifference curve. The two have the same shape, but output can be counted in books and satisfaction cannot be counted. A larger output on a higher isoquant is a fact, not an opinion.

IN HISTORY

In 1769 Richard Arkwright, an English inventor and businessman, patented a spinning machine that later became known as the water frame. He first set up a horse-powered mill in Nottingham, a city in the English Midlands, to run his machines.

Horses gave too little power, so in 1771 Arkwright built a mill at Cromford in Derbyshire. England's first

The main ideas of the chapter in a few lines.

A common belief is that the firm should use the input with the lower price; the rule compares output per taka.

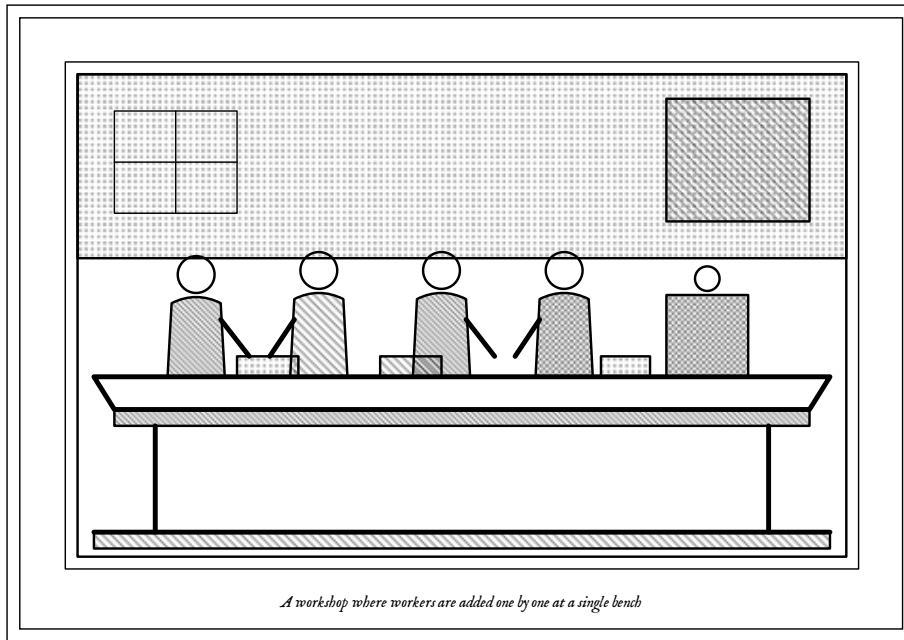
A second belief is that least cost means fewest inputs; it means the lowest total bill.

A third belief is that the best mix spends the same amount on each input; it does not.

A fourth mistake is to read an isoquant like an indifference curve with measurable satisfaction.

In 1769 Richard Arkwright patented a spinning machine and ran it first in a horse-powered mill in Nottingham.

In 1771 he built Cromford Mill in Derbyshire, the first



CHAPTER X

Production Function and Returns

Why does the sixth worker add less than the second?



This chapter shows how inputs become output, why each extra worker added to a fixed workshop eventually adds less than the one before, and what happens to output when every input grows together.

A rise in the wage turns the isocost line steeper and moves the best mix toward machines.

Recap: the least-cost rule in two forms and the effect of a wage change.

* * *

12.3 USING THE RULE

What happens when the firm wants to make more than 100 books?

The expansion path joins the least-cost combinations for every output.

Cost-minimising asks how to make a given output; profit-maximising also asks how much to make.

In the short run some inputs are fixed, so the firm may be unable to reach the least-cost combination.

The rule assumes two inputs, steady prices and smooth substitution, and real firms do so.

marginal product per taka: marginal product of a worker / wage = marginal product of a machine / machine rent. If workers add more output per taka than machines, the firm should hire more workers and rent fewer machines. As it does so, the marginal product of a worker falls and that of a machine rises, until the two ratios are equal.

Now suppose the wage rises while the machine rent stays the same. The isocost line turns steeper, because a worker now costs as much as more machines. The point of contact moves along the isoquant toward fewer workers and more machines. The firm replaces workers with machines, because machines have become cheaper by comparison. Total cost for the same output still rises, but by less than it would if the firm kept the old mix. The worked examples give numbers.

Before moving on. The least-cost combination is found where the lowest isocost line touches the isoquant. There, MRTS equals the price ratio, which is the same as equal marginal product per taka. A rise in the wage steepens the isocost line and moves the best mix toward machines.

Suppose the firm wants to make more than 100 books a day. The same method applies at each output. Each output has its own isoquant, and a larger output has an isoquant farther from the corner. For each one, the lowest isocost line that touches it gives a new least-cost combination.

If we join these least-cost points, the line is called the *expansion path*. It shows how the best mix changes as output grows, while the input prices stay the same. Usually the path rises, because a larger output needs more of both inputs. The total cost at each point on the path is the least total cost of that output, which is the total cost schedule used in Chapter 11, 'Costs of Production'.

Two questions must be kept apart. A cost-minimising choice asks: for a given output, which mix costs least? A profit-maximising choice asks a larger question: which output should the firm make? A firm that wants the largest profit must answer both. It first finds the cheapest way to make each output, and then it chooses the output that gives the best gap between revenue and total cost. A firm can minimise cost and still lose money if it chooses a poor output. Later chapters deal with that second choice.

In the short run, some inputs cannot be changed, and in the long run all can. Suppose, as an illustration, that a firm has signed a contract for 6 machines. For 100 books it must use the mix (3, 6), which costs Tk 4,200 in the case. The least cost is Tk 4,000, so the contract adds Tk 200 a day. Only in the long run can the firm move to (4, 4).

The rule has limits. It assumes two inputs, steady prices and smooth substitution between them. Some inputs fit together in fixed amounts: a bus needs one driver, however cheap the fuel.

Change the combination, and the output changes. Two helpers and two ovens might give 180 loaves. One helper and one oven might give 60. The production function lists all such pairs. It does not say which combination to choose, because that depends on prices and costs, which later chapters study. It states only what is technically possible, and it assumes the best known methods.

Before moving on. Inputs are what a firm uses, and output is what it makes. A production function is the rule that shows the largest output for each combination of inputs. It describes what is possible, not what is wise.

THE SHORT RUN AND THE LONG RUN

Not every input can be changed at once. A baker can hire another helper by next week, but a new oven must be ordered, delivered and installed, which may take months. A firm cannot adjust all its inputs at the same speed, and the speed decides what choices it has.

The **short run** means a period so short that at least one input, such as a machine, cannot be changed. In the short run the baker may hire more helpers and buy more flour, but the oven stays. An input that cannot be changed in the period is called a fixed input, and an input that can be changed is a variable input. The short run is not a number of days or months. It is whatever period leaves at least one input fixed.

The **long run** means a period long enough for all inputs to be changed. Given enough time, the baker can build a new bakery, buy five ovens and move to another street. In the long run nothing is fixed. Different firms have different long runs: a roadside tea stall can change every input in a week, while a steel plant may need years. The two periods divide this chapter, because the law in section 10.2 belongs to the short run, and the idea in section 10.3 belongs to the long run.

Before moving on. In the short run at least one input is fixed. In the long run all inputs can be changed. The length of each depends on the firm and the industry.

TOTAL, MARGINAL AND AVERAGE PRODUCT

We now need three measures of output. They are needed to follow how output changes when the firm varies one input, such as workers, while all other inputs stay fixed. Take the smallest possible example, a baker with one oven. With 1 helper the baker makes 6 loaves a day, and with 2 helpers 14 loaves a day.

The first measure is the simplest. **Total product** means the whole quantity of output produced with a given amount of inputs. With 1 helper, total product is 6 loaves. With 2 helpers, it is 14 loaves. It is the number that the firm sells.

The second measure asks what the last helper added. **Marginal product** means the extra output from adding one more unit of an input, others held fixed. In words: marginal product equals the change in total product divided by the change in the input. In symbols, $MP = \Delta TP \div \Delta L$, where MP stands for marginal product, TP for

Product. Word in use. In daily life, a good that is offered for sale. In this chapter, the quantity of output, as in total product.

Different combinations of inputs give different outputs.

Before moving on: a production function ties inputs to the largest output.

Some inputs can be changed quickly, and others cannot.

Short run. A period so short that at least one input, such as a machine, cannot be changed.

The short run is a period in which at least one input is fixed.

Long run. A period long enough for all inputs to be changed.

The long run is a period in which every input can be changed.

Before moving on: the short run has a fixed input and the long run has none.

GO SLOWLY HERE.

Three measures describe how output responds to one input.

Total product. The whole quantity of output produced with a given amount of inputs.

Total product is the whole output from the given inputs.

Marginal product. The extra output from adding one more unit of an input, others held fixed.

Marginal product is the extra output from one more unit of

added and the MRTS, which is the machines saved divided by the workers added.

MRTS BETWEEN NEIGHBOURING MIXES ON THE ISOQUANT FOR 100 BOOKS

MOVE	MACHINES SAVED	WORKERS ADDED	MRTS
(2, 10) to (3, 6)	4	1	4
(3, 6) to (4, 4)	2	1	2
(4, 4) to (5, 3)	1	1	1
(5, 3) to (6, 2)	1	1	1

Read the first row: adding one worker saves four machines, so the MRTS is 4. The values fall as we move down the table. This is called a diminishing MRTS. It happens because the more workers there are, the less each extra worker adds, while the fewer machines remain, the more each remaining machine matters. The falling MRTS is the reason the isoquant curves toward the corner.

There is a second way to write the MRTS. The marginal product of an input is the extra output from one more unit of it, a quantity met in Chapter 10, 'Production Function and Returns'. A worker replaces machines in proportion to what each adds, so $MRTS = \text{marginal product of a worker} / \text{marginal product of a machine}$. A worker who adds twice as much output as a machine replaces two machines.

Before moving on. An isoquant joins the mixes of two inputs that give the same output. An isocost line joins the mixes that have the same total cost, and its slope is the price ratio. The MRTS is the number of machines saved by adding one worker, and it falls as workers replace machines.

* * *

12.2 THE LEAST-COST RULE

Return to the workshop. Five mixes give 100 books. Which of them costs least? The method has three steps. First, list the mixes on the isoquant. Second, find the total cost of each mix. Third, choose the mix with the smallest total cost.

The result has a name. The **least-cost combination** is the mix of inputs that produces a given output at the lowest total cost. The words *given output* matter. The firm does not choose how much to make here. It has fixed the output, and it asks only how to make it most cheaply.

Case file, illustrative. For an output of 100 books bound a day, with the wage at Tk 600 and the machine rent at Tk 400, the total costs are: (2, 10) = Tk 5,200; (3, 6) = Tk 4,200; (4, 4) = Tk 4,000; (5, 3) = Tk 4,200; (6, 2) = Tk 4,400. The least cost is (4, 4) at Tk 4,000. The MRTS moves from 2 to 1 around the price ratio $600 / 400 = 1.5$.

The table shows how each total is built. The wage bill is Tk 600 times the number of workers, and the rent bill is Tk 400 times the number of machines. The last column adds the two.

The MRTS falls as workers replace machines, because each added worker is worth less.

The MRTS also equals the marginal product of a worker divided by that of a machine.

Recap: inputs, the isoquant, the isocost line and the MRTS.

Of five mixes that make 100 books, which costs least?

Least-cost combination. The mix of inputs that produces a given output at the lowest total cost.

The least-cost combination is the cheapest mix for a given output.

Case file: the cheapest of five mixes is 4 workers and 4 machines at Tk 4,000.

The table splits each total cost into a wage bill and a rent bill.

IN SHORT

Hold some inputs fixed, keep adding one input, and the extra output from each new unit will eventually become smaller and smaller.

PAUSE AND CHECK 10.P3 State the law of diminishing returns in your own words and name the fixed input in the binding workshop.

WHY RETURNS DIMINISH, AND WHEN THE LAW HOLDS

Why does the pattern appear? In the binding workshop the machine is fixed. The first worker must do every task alone, moving between folding, sewing and covering, and so wastes time. The second worker allows each person to specialise, and the machine is kept busy. This is why marginal product rises at first. Soon the machine is in use all day. Each new worker now has less machine time to share, must wait for a turn, or takes over a task someone else was doing well. By the sixth worker the machine can produce no more, and no addition of labour can change that.

The law holds under conditions that you should state each time. First, at least one input is fixed, so the law is a short-run law. Second, the methods of production stay the same; a better machine would move the whole schedule up. Third, the units of the variable input are alike. The workers in the workshop are equally skilled, so the fall in marginal product is not caused by hiring worse workers. It is caused by the crowding of a fixed input. If a student explains the fall by saying 'the later workers are lazier', the explanation is wrong, and the law is lost.

A common error concerns what diminishes. Total product does not have to fall, and in our case it rises until the fifth worker. What diminishes is the marginal product, the addition to output. An example from farming helps. Put one more labourer on a single field, and the harvest rises. Put a hundred more, and it hardly rises, because the soil, not the labour, limits the harvest. The extra output shrinks long before any harm is done.

To read the first figure, look at the axes. The horizontal axis shows the number of workers; the vertical axis shows total product. Each point shows the output for one number of workers. The curve rises steeply for the first two workers, flattens through the fourth and fifth, and levels out at the sixth, where it no longer rises.

The cause is a fixed input shared among more and more workers.

The law holds only under stated conditions.

The law is about the extra output, not about the whole output.

The first figure shows total product rising, quickly at first, then slowly, then not at all.

YOU WILL NEED

Chapter 10 'Production Function and Returns', section 10.1 'Inputs and Output'; Chapter 11 'Costs of Production', section 11.1 'Explicit and Implicit Costs'; Chapter 7 'Indifference Curve Analysis', section 7.2 'Marginal Rate of Substitution'; Appendix A1 'Maths You Will Use'.

A workshop owner must bind 100 books every day and can use workers, machines or both.

Each way has its own daily bill, so the owner needs a method to find the smallest.

Trying every way is slow and explains nothing; two lines give a better method.

Recall: production function, Chapter 10, section 10.1 'Inputs and Output'.

A firm makes output from inputs, and a mix of inputs means the amounts used together.

Input. Word in use. In daily life, a contribution of ideas or advice, as in your input to a meeting. In economics, a thing a firm uses to make output, such as labour, a machine or a raw material.

Mix. Word in use. In daily life, to blend things together. In economics, the amounts of different inputs that a firm uses together, as in a mix of workers and machines.

Workers and machines can replace each other within limits, so many mixes give one output.

Imagine the owner of a binding workshop who must bind 100 books every day. The books can be bound by hand, by machine, or by some of each. With hand work only, the workshop needs many workers and no machine. With machines only, it needs many machines and few workers. Between these two extremes lie many other ways to reach the same 100 books.

Each way has its own bill at the end of the day. Workers must be paid, and machines must be rented. More machines mean fewer workers, so one bill rises as the other falls. The owner is left with a question that has a definite answer: which way gives the same 100 books for the smallest total?

A beginner might try every way, one after another. That works when there are few ways, but it does not explain why one way wins. A better method uses two pictures. One line shows every way of making the output. Another line shows every way of spending a given sum. Where the two lines meet in the right manner, the cheapest mix is found. Many students find this idea new, but the thinking is familiar from the consumer's choice in Chapter 7, now applied to a firm. We start with the first line.

* * *

12.1 CHOOSING INPUTS

A firm makes its output by using inputs. Chapter 10 showed that a production function links the inputs to the output. In this chapter we study two inputs: workers and machines. Other inputs, such as paper and glue, are held fixed. Four workers with four machines is one mix. Five workers with three machines is another mix.

Workers and machines can replace each other, within limits. A machine can do the work of several people, and a skilled worker can do what a poor machine cannot. So the same output can come

Average product is rising, so each worker added makes the others more productive. The second stage runs from that point to the point where marginal product reaches zero, here the sixth worker. Marginal product is positive but falling, and average product is falling. The third stage begins when marginal product turns negative, which means total product falls. The binding workshop has not reached it.

A careful firm avoids the third stage, since no firm wants to pay for a worker whose presence lowers output. It also has little reason to stop in the first stage, since extra workers raise the average output of all. This reasoning points a firm toward the second stage. Where in the second stage the owner stops depends on costs and prices, which Chapter 11 and later chapters study.

Before moving on. The law of diminishing returns says that with some inputs fixed, extra units of one input eventually add less and less. Marginal product rises at first, then falls; average product rises while marginal product is above it. The three stages divide the schedule at the peak of average product and at the point where marginal product reaches zero.

* * *

10.3 RETURNS TO SCALE

CHANGING EVERY INPUT

The law of diminishing returns changed one input and held the others fixed. The owner can ask a different question: if the workshop is rebuilt with twice as many workers and twice as many machines, will output double? The answer matters for the long run, when all inputs can change.

Returns to scale means the change in output when all inputs are increased in the same proportion. Double the workers and double the machines, and see what happens to output. There are three possible answers, and each has a name. The test is to compare the proportional rise in output with the proportional rise in inputs.

Output may rise by more than the inputs. This is increasing returns to scale. It may rise by exactly as much, which is constant returns to scale. Or it may rise by less, which is decreasing returns to scale. In symbols, if inputs are multiplied by k , output is multiplied by more than k , by exactly k , or by less than k . Here k stands for any number above 1, such as 2 or 1.5.

Case file, illustrative. In the binding workshop, 2 workers and 1 machine give an output of 24. With 4 workers and 2 machines the output is 60. With 6 workers and 3 machines it is 90. With 8 workers and 4 machines it is 108.

A firm usually works in the second stage.

Before moving on: the law, the schedule and the stages describe one pattern.

GO SLOWLY HERE.

The long run raises every input together, which is a different question.

Scale. Word in use. In daily life, an instrument for weighing, or the ratio on a map. In economics, the size of a firm's operation, meaning the amount of all its inputs taken together.

Returns to scale. The change in output when all inputs are increased in the same proportion.

Returns to scale is the change in output when all inputs grow in the same proportion.

The three cases are compared by the size of the rise in output.

The record gives four outputs for four sizes of workshop.

Higher output spread fixed costs and lowered cost per car, as the falling part of the U predicts.

The usual account is that the cost of making each car fell as output rose, and Ford passed much of the saving to buyers as lower prices. The figures above are prices, not costs. The fall in cost per car is the falling part of the average total cost curve, where large fixed costs are shared among very many units.

* * *

REVIEW QUESTIONS

A print stall has a fixed cost of Tk 200 a day. At outputs of 0, 5, 10, 15, 20, 25 and 30 pages of posters a day, its variable cost is Tk 0, 100, 180, 240, 320, 430 and 560. Use these data for questions 11.2 to 11.6.

11.1 A tailor spends Tk 8,000 on cloth and Tk 3,000 on a helper's wage in a month. She works in her own shop, and her time is worth Tk 12,000 a month elsewhere. The building she owns could be rented for Tk 5,000. Find her explicit costs and her implicit costs.

11.2 Find the total cost of the print stall at each output from 0 to 30.

11.3 Find the marginal cost of each page in every step of five pages.

11.4 Find average total cost at each output from 5 to 30, to two decimal places. At which output is it lowest?

11.5 The stall works 6 days a week and makes 20 pages every day. What is its total cost for the week?

11.6 The stall's daily fixed cost rises from Tk 200 to Tk 260. State what happens to total cost at 20 pages, to marginal cost, and to average total cost at 20 pages.

11.7 A shopkeeper's accounts show a profit of Tk 500,000. Her own time is worth Tk 360,000 a year elsewhere. Her capital of Tk 3,000,000 could earn 8 per cent a year. Find her implicit costs and the result after counting them.

dozen. Seventh: $3.75 - 4 = -0.25$ dozen, and $-0.25 \times 12 = -3$ items. *Answer:* the sixth adds nothing and the seventh takes away 3 items. *Check:* a negative marginal product means total product is falling: the third stage.

Given: a bakery with 1 oven and 2 helpers makes 30 loaves a day. With 2 ovens and 4 helpers it makes 66. With 2 ovens and 5 helpers it makes 76. *Find:* the returns to scale from the first to the second combination, and the marginal product of the fifth helper. *Steps.* Inputs are multiplied by 2. Output is multiplied by $66 \div 30 = 2.2$. The fifth helper's marginal product, with the ovens fixed at 2, is $76 - 66 = 10$. *Answer:* increasing returns to scale, and a marginal product of 10 loaves. *Check:* the first answer changed all inputs; the second changed one input only.

* * *

SUMMARY

A production function shows the largest output from each combination of inputs. In the short run at least one input is fixed; in the long run none is. Total product is the whole output, marginal product is the extra output from one more unit, and average product is output per unit. The law of diminishing returns says that, with some inputs fixed, extra units of one input eventually add less and less. Returns to scale asks what happens when all inputs rise in the same proportion.

THINK IT THROUGH

10.T1 The sixth worker in the binding workshop adds nothing. Explain why a seventh worker is unlikely to add anything, without using the idea that later workers are less able.

10.T2 The workshop pays each worker the same daily wage. Why does the labour cost of each extra unit of output rise as more workers are hired?

WHERE PEOPLE GO WRONG

Many students think that diminishing returns means output falls; it means that the extra output shrinks. Some explain it by saying that later workers are less skilled, when the cause is the crowding of a fixed input. Others confuse the law with returns to scale, and the table in section 10.3 shows why they differ. The last error is to treat marginal and average product as the same measure; the first is the last unit's addition, and the second is the output per unit.

IN HISTORY

Thomas Robert Malthus (1766 to 1834) was an English clergyman and scholar. In 1798 his book *An Essay on the Principle of Population* appeared in London, first without the author's name. He wrote it to make the harmful idea of the earlier William Godwin. The

Rung 4: a combined problem uses the law and returns to scale together.

The chapter's main results in a few lines.

Four common misunderstandings, each corrected.

In 1798 Malthus applied the idea of shrinking extra output to food.

able to say why the bottom of the U is where marginal cost equals average total cost.

* * *

WORKED EXAMPLE: COSTS OF THE BINDING WORKSHOP

We now work two examples in the same order each time: Given, Find, Formula, Steps, Answer and Check. The first uses the workshop table at an output of 30 books. The second uses the shop's own records and shows how implicit costs change the picture.

Given. Fixed cost is Tk 600 a day. Variable cost is Tk 520 at 20 books and Tk 720 at 30 books. *Find.* Total cost at 30 books, marginal cost of each book from 20 to 30 books, and average total cost at 30 books. *Formula.* In words, total cost is fixed cost plus variable cost. In symbols, $TC = FC + VC$. Marginal cost is $MC = (\text{change in } TC) \div (\text{change in } Q)$, and average total cost is $ATC = TC \div Q$.

Steps. Total cost at 30 books: $TC = 600 + 720 = \text{Tk } 1,320$. Total cost at 20 books: $TC = 600 + 520 = \text{Tk } 1,120$. Change in $TC = 1,320 - 1,120 = \text{Tk } 200$. Change in $Q = 30 - 20 = 10$ books. Marginal cost: $MC = 200 \div 10 = \text{Tk } 20$ a book. Average total cost: $ATC = 1,320 \div 30 = \text{Tk } 44$ a book.

Answer. At 30 books a day, total cost is Tk 1,320, the marginal cost of each book from 20 to 30 is Tk 20, and average total cost is Tk 44 a book. *Check.* Multiply back: $44 \times 30 = 1,320$, which equals total cost. The marginal cost of Tk 20 is below the average of Tk 44, so the average should be falling, and the table shows it falling from Tk 56 to Tk 44. Both figures agree with the table.

Given. For 2025, the profit shown in the accounts of Meghna and Sons is Tk 750,000.⁴ The owner's time is valued at Tk 600,000. Capital employed is Tk 4,100,000, and the interest rate the capital could earn elsewhere is 7 per cent a year. *Find.* The implicit costs, and the result after counting them. *Formula.* Forgone interest = rate $\times$ capital employed. Result after implicit costs = profit shown in the accounts - total implicit costs.

Steps. Write 7 per cent as $7 \div 100 = 0.07$. Forgone interest = $0.07 \times 4,100,000 = \text{Tk } 287,000$. Total implicit costs = $600,000 + 287,000 = \text{Tk } 887,000$. Result = $750,000 - 887,000 = -\text{Tk } 137,000$.

Answer. The implicit costs are Tk 887,000, and after counting them the shop falls Tk 137,000 short. *Check.* The implicit costs are larger than Tk 750,000, so the result must be negative. Adding back: $-137,000 + 887,000 = 750,000$, the figure in the accounts. What does the owner learn? Her resources would have earned Tk 137,000 more elsewhere, if other things were equal. She may still choose to carry on, because she may value the shop for other reasons. The calculation tells her the price of that choice.

Common slips are easy to name. First, forgetting fixed cost when adding up total cost. Second, dividing by the output when you need the change in output, or the reverse. Third, forgetting units, ⁴ Profit has several exact meanings, taught in Chapter 16, 'Interest and Profit'. Here it means only the money left after explicit costs are paid.

Two examples apply the formulas, one small and clean and one realistic.

The first example gives total, marginal and average total cost at 30 books.

Each line of arithmetic is shown, with its unit, so the method is the only new thing.

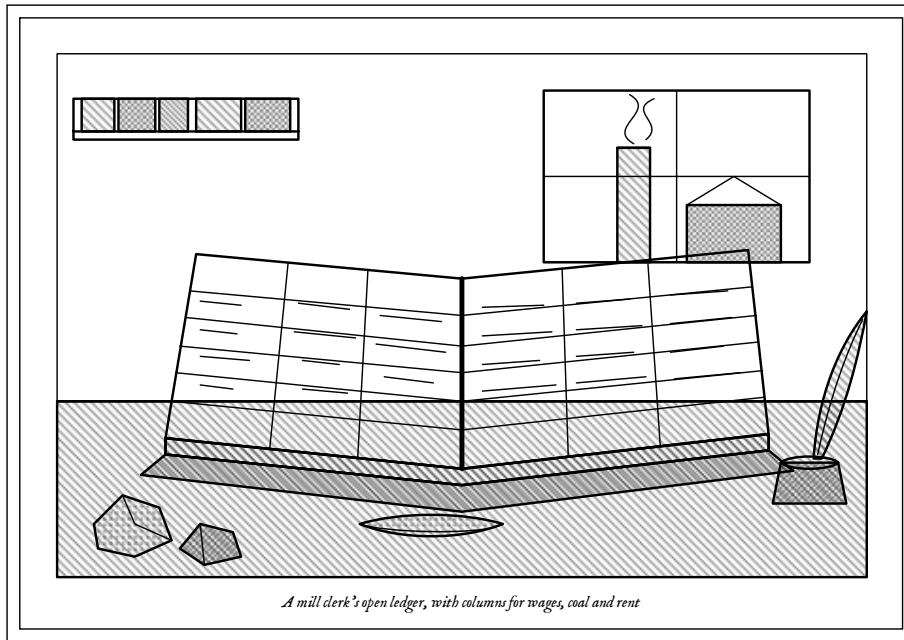
The answer is stated in a sentence and tested against the table.

The second example uses the shop's records to count implicit costs.

Every step is written out, so that nothing depends on guesswork.

The shop looks successful in its accounts but falls short once its own resources are valued.

Five common slips spoil these calculations, and each has a simple remedy.



CHAPTER XI

Costs of Production

Why can a shop pay every bill and still lose?



Rent, wages and paper are easy to count. Other costs hide, because nobody sends a bill for them. This chapter sorts costs into kinds, shows how to calculate each one by hand, and explains why the cost curves take the shapes they do.

The section ends with a recap and a test of understanding.

A cost curve draws a table as a picture, with one point for each output.

GO SLOWLY HERE.

IN SHORT

Total cost equals fixed cost plus variable cost. Marginal cost is the change in total cost divided by the change in output. Average total cost is total cost divided by output.

* * *

11.3 THE COST CURVES

A curve shows a table at a glance. The first figure below draws the workshop's fixed, variable and total cost. Here is how to read it. The horizontal axis, the line along the bottom, shows output in books a day. The vertical axis, the line up the side, shows cost in taka a day. Each point on a curve tells you the cost at one output. Start at an output on the bottom, move upward until you meet a curve, and read the cost from the side.

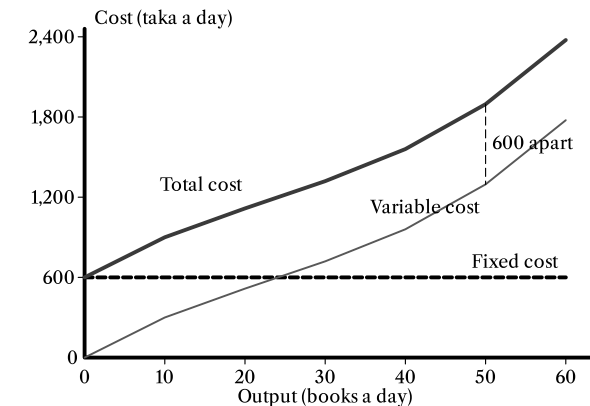


FIG. 11.1. Fixed, variable and total cost curves of the binding workshop

The fixed cost line is flat, because fixed cost does not change with output. The variable cost curve starts at zero, because nothing is spent on variable inputs when nothing is made. The total cost curve has exactly the shape of the variable cost curve, but it sits Tk 600 higher at every output. That gap is the fixed cost. At 30 books, for example, variable cost is 720 and total cost is 1,320.

The variable cost curve is gentle at first and steep later. The reason is the law of diminishing returns, which says that adding more of one input to fixed amounts of others will, after a point,

³ Chapter 10, 'Production Function and Returns', section 10.2 'The Law of Diminishing Returns', explains why the extra output from each new worker rises at first and then falls.

The second kind is easy to miss. An **implicit cost** is the value of the resources that the owner supplies to the firm, for which no payment is made. The word *implicit* means present but not openly stated. For example, an owner who works in her own shop without a salary supplies her time. That time has a value, because she could sell it to another firm. In the same way, money the owner has placed in the firm could have earned interest at a bank. Each of these has an opportunity cost, though nobody sends a bill.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna. Record for 2025: sales Tk 14,400,000; profit shown in the accounts Tk 750,000; capital employed Tk 4,100,000; rent Tk 60,000 a month; 7 staff with total wages Tk 2,940,000; bank loan Tk 600,000. Rent, wages and interest on the loan are explicit costs, and each appears in the shop's records. The case file values the owner's time at Tk 600,000 for the year. It also records forgone interest of 7 per cent on the capital employed, which is Tk 287,000. These two amounts are implicit costs. No bank record shows either one.

The test for an implicit cost is one question: what is the best use of this resource outside the firm? The answer gives its value. Nurul Meghna's time could earn a salary elsewhere, and her capital could earn interest. Each answer is the opportunity cost of one of her own resources. Many students find this step confusing, because the owner never feels the sacrifice as a payment. The confusion disappears when you ask the same question about every resource, whoever owns it. A rented room has an explicit cost. An owned room has an implicit cost. Both are costs.

The viewpoint changes the list. The accountant records explicit costs only. The owner, who must decide whether to carry on, counts both kinds, and economists take the owner's view. There are limits. An implicit cost is an estimate, not a receipt. Also, a resource with no other use has no implicit cost. This answers the puzzle at the start. Bina's time had no better use, so her implicit cost is nil, while Asha's is Tk 1,100 a day. Her stall leaves Tk 900 after paid costs, which is Tk 200 less than her old wage.

Before moving on. A cost is either explicit, involving a payment to outsiders, or implicit, the value of what the owner supplies without payment. We find an implicit cost by asking what the resource could earn elsewhere. The accountant records explicit costs only, and the economist counts both kinds. You should now be able to say, in your own words, why an owner's unpaid time is a cost.

PAUSE AND CHECK 11.P1 A tailor owns the building of her shop and could rent it to another person for Tk 5,000 a month. She pays rent to nobody. Is there a rent cost, and if so, of which kind?

* * *

11.2 SHORT-RUN COSTS

The second kind of cost is the value of what the owner supplies without being paid.

Implicit cost. The value of resources the owner supplies, which involve no payment but have an opportunity cost.

The shop's records show explicit costs openly and implicit costs not at all.

To find an implicit cost, ask what the owner's own resource could earn elsewhere.

The accountant counts payments, while the owner counts everything given up.

The section closes with a recap and a test of understanding.

Costs are studied in the short run, when at least one input cannot be changed.

A fixed cost stays the same whatever the output.

Fixed cost. A cost that does not change with the quantity produced in the short run, such as rent.

A variable cost rises and falls with output.

Variable cost. A cost that changes with the quantity produced, such as paper and wages for extra output.

Total cost joins the two kinds, so it is never smaller than the fixed cost.

Total cost. Fixed cost plus variable cost at a given output.

The three costs fit together in one short sum.

The workshop's records give fixed, variable and total cost at seven levels of output.

To read a row, add the fixed and variable columns to get the total.

All the costs in this section belong to the short run. The short run is a period so short that at least one input, such as a machine, cannot be changed.² In the short run, therefore, the firm cannot avoid paying for its machine, however little it makes. Inputs that can be changed quickly, such as paper and extra workers, give the firm room to raise or cut its output.

The stall fee is the model. A **fixed cost** is a cost that does not change with the quantity produced in the short run, such as rent. A workshop's machine rent is fixed: it is owed on a day when ten books are bound and on a day when sixty are bound. Fixed cost is the cost of being ready to produce, not of producing. It is still a cost, and it must be paid.

The milk, tea leaves and sugar are the model for the other kind. A **variable cost** is a cost that changes with the quantity produced, such as paper and wages for extra output. If the stall sells more cups, it buys more milk. If it sells none, it buys none. In a workshop, paper, thread, glue and the wages of extra workers all rise as more books are bound.

Total cost is fixed cost plus variable cost at a given output. In symbols, $TC = FC + VC$, where TC is total cost, FC is fixed cost and VC is variable cost, all measured in taka for the same period. The tea stall at 10 cups has $TC = 100 + 40 = \text{Tk } 140$. Even at zero output, total cost equals fixed cost, because the fee is still owed.

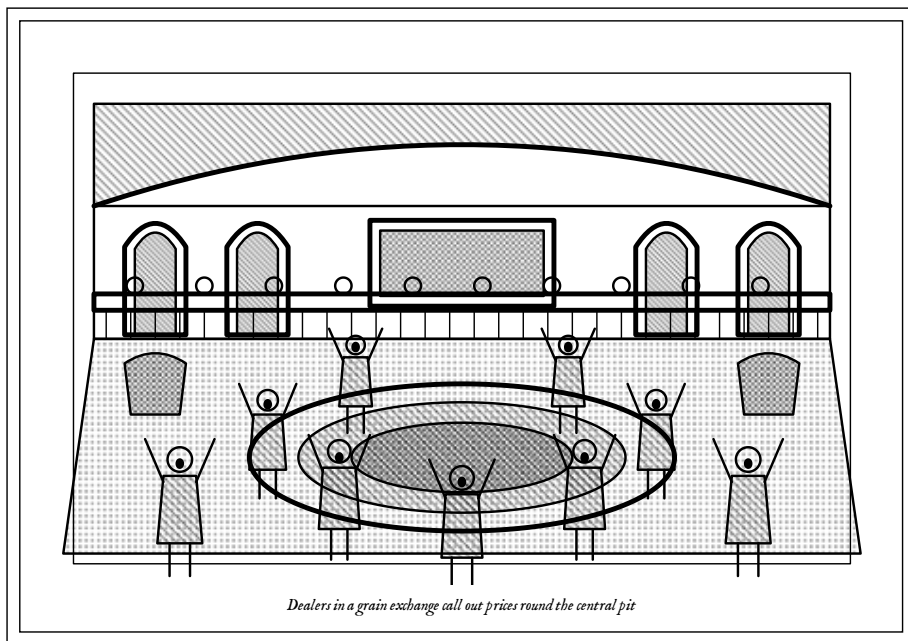
Before moving on. Fixed cost does not move with output. Variable cost moves with output. Total cost is the two added together. You should now be able to say why total cost at zero output is not zero.

Case file, illustrative. A binding workshop uses one machine and a small team to bind books. Its fixed cost is Tk 600 a day. At daily outputs of 0, 10, 20, 30, 40, 50 and 60 books, its variable cost is Tk 0, 300, 520, 720, 960, 1,300 and 1,780. Adding the fixed cost to each variable cost gives total cost of Tk 600, 900, 1,120, 1,320, 1,560, 1,900 and 2,380. The table below sets these records in columns.

COSTS OF THE BINDING WORKSHOP, IN TAKA A DAY

OUTPUT (BOOKS A DAY)	FIXED COST	VARIABLE COST	TOTAL COST
0	600	0	600
10	600	300	900
20	600	520	1,120
30	600	720	1,320
40	600	960	1,560
50	600	1,300	1,900
60	600	1,780	2,380

Read each row from left to right. In the row for 20 books, fixed cost 600 plus variable cost 520 gives total cost 1,120. Notice that the fixed column never changes, while the variable column climbs faster and faster. Notice also that the first row shows an output of 0.² Chapter 10, 'Production Function and Returns', section 10.1 'Inputs and Output', defines the short run and the long run.



CHAPTER XIII

Market Structure and Perfect Competition

Why must a grain dealer accept a price that the dealer did not choose?



This chapter sorts markets into four kinds by the number of sellers, the sameness of the good and the ease of entry. It then studies the firm in the most competitive kind of market, finds the output that gives that firm the largest profit, and follows the profit through time.

quality on offer. Each feature changes how much power a seller holds over the price.

The word *competition* needs care. In daily life it means a contest, and a seller who advertises loudly is said to compete hard. In economics the word is narrower. Economists judge competition by the number of rivals and the ease of entry, and not by the loudness of advertising. A market can contain fierce advertising and weak competition at the same time, for example when only two firms exist and no third firm may enter.

The next table sorts markets into four kinds. Read a row from the left: the first column names the kind, and the other columns give the number of sellers, the kind of good and the ease of entry. The last three kinds are taught in Chapter 14, 'Monopoly and Imperfect Competition'; the section on monopoly is 14.1 'Monopoly'. Here we use the table only to place the first kind among the others.

FOUR KINDS OF MARKET

KIND	SELLERS	GOOD	ENTRY
Perfect competition	Very many	Identical	Free
Monopolistic competition	Many	Different versions	Easy
Oligopoly	A few	Identical or different	Difficult
Monopoly	One	One good with no close substitute	Blocked

Markets and firms need a link word between them. The **industry** means all the firms that produce the same kind of good. All the firms that make notebooks form the notebook industry; all the wheat farmers of a country form the wheat industry. The quantity supplied by the industry is the sum of the quantities supplied by its firms. The difference between one firm and its industry matters greatly in section 13.3, where firms enter and leave.

The table has limits. Real markets rarely match one row exactly, and a market with five large firms and fifty small ones belongs partly to two rows. A map leaves out detail so that the main roads are easy to see. Use the table to ask which features a market has.

Before moving on. A market structure is the set of features that shape how firms behave: number of sellers, kind of good, ease of entry and information. There are four kinds of market. An industry is all the firms that produce the same kind of good.

* * *

13.2 THE PERFECTLY COMPETITIVE FIRM

Imagine a hall in which dozens of dealers trade the same grain. One price is called across the floor. A dealer who asks for more finds no

Competition. Word in use. In daily life, a contest in which each person tries to win, such as a race. In economics, the pressure that sellers place on one another through price and quality when several offer the same good.

Competition has an everyday meaning and a narrower meaning in economics.

The table gives four kinds of market, sorted by the features above.

Industry. All the firms that produce the same kind of good.

We also need one word for all the firms that make the same good.

Real markets rarely match one row exactly, so the table is a map and not a photograph.

Before moving on, recall the three ideas of section 13.1.

A hook: in a grain hall, one price is called across the floor and every dealer must accept

WORKED EXAMPLES: FOUR RUNGS

Each example below follows the order Given, Find, Formula, Steps, Answer, Check.

Given: a nominal wage of Tk 20,000 a month, and a price index that rises from 100 to 125. *Find:* the real wage. *Formula:* real wage = nominal wage ÷ (price index ÷ 100). *Steps.* 125 ÷ 100 = 1.25, and 20,000 ÷ 1.25 = 16,000. *Answer:* Tk 16,000 a month in base-year prices. *Check:* prices rose by a quarter, so the wage should buy only four-fifths as much: 20,000 × 0.8 = 16,000.

Given: three shop sites have profits before rent of Tk 1,470,000, Tk 1,100,000 and Tk 750,000 a year, and the last is the marginal site. *Find:* the rent that each site can bear. *Formula:* rent = profit before rent at the site - profit before rent at the marginal site. *Steps.* First site: 1,470,000 - 750,000 = 720,000. Second site: 1,100,000 - 750,000 = 350,000. Marginal site: 0. *Answer:* Tk 720,000, Tk 350,000 and nothing. *Check:* the better the site, the higher the rent, and the first figure agrees with the Dhanmondi record.

Given: an annual nominal wage of Tk 420,000, and a price index that rises from 100 to 112. *Find:* the real wage per month. *Steps.* The monthly nominal wage is 420,000 ÷ 12 = 35,000. The real monthly wage is 35,000 ÷ 1.12 = 31,250. *Answer:* Tk 31,250 a month. *Check:* the annual real wage is 420,000 ÷ 1.12 = 375,000, and 375,000 ÷ 12 = 31,250, the same. The slip to avoid is to divide by 12 at one stage and forget it at another.

Given: a firm pays Tk 35,000 a month. Successive extra workers add Tk 52,000, Tk 41,000, Tk 36,000 and Tk 30,000 a month of revenue. The third worker would have accepted Tk 28,000 a month. *Find:* how many to hire, the gain, and the third worker's economic rent. *Steps.* Compare each marginal revenue product with 35,000: the first three exceed it, the fourth (30,000) does not, so hire three. The revenue added is 52,000 + 41,000 + 36,000 = 129,000, and the wages are 3 × 35,000 = 105,000, a gain of 24,000. The economic rent of the third worker is 35,000 - 28,000 = 7,000. *Answer:* hire three, gain Tk 24,000 a month, and the third worker's economic rent is Tk 7,000. *Check:* the last worker hired adds 36,000, slightly above the wage of 35,000, as the rule requires.

* * *

SUMMARY

Rent is the payment to the owner of land for its use. Land as a whole is fixed in supply, so its whole payment is economic rent, which is any payment above the least needed to keep a factor in use. Ricardo's theory makes rent the difference between a plot's yield and that of the marginal land. Quasi-rent is a temporary surplus earned by a man-made asset in the short run. A wage is the payment to labour, the nominal wage is in money and the real wage is in goods. A firm hires while the marginal revenue product is at least the wage, and wages are set by demand and supply.

Four examples, from clean numbers to a combined problem.

Rung 1: a real wage with clean numbers.

Rung 2: differential rent with three sites and less tidy numbers.

Rung 3: the period changes from a year to a month.

Rung 4: hiring and economic rent in one problem.

The chapter's main results in a few lines.

profit. So the firm raises its output until marginal cost reaches the price. This is the rule *marginal cost equals price*.

Case file, illustrative. The market price is Tk 40, taken from the notebook market. Cost data of the binding workshop (Chapter 11): marginal cost for each 10-unit step is 30, 22, 20, 24, 34 and 48, at outputs 10 to 60. Each figure is the extra cost of one unit in that step. A step of 10 units therefore adds 10 times its marginal cost to total cost.

The table below applies the rule one step at a time. Read a row from the left: the first column names the step by its last unit, the second gives the extra revenue (10 units at Tk 40), the third gives the extra cost (10 times the marginal cost), and the fourth gives the difference. A positive difference means the step adds to profit.

EACH 10-UNIT STEP AT A PRICE OF TK 40

STEP ENDING AT OUTPUT	EXTRA REVENUE	EXTRA COST	EXTRA PROFIT
10	400	300	+100
20	400	220	+180
30	400	200	+200
40	400	240	+160
50	400	340	+60
60	400	480	-80

Now read the second chart. The horizontal axis shows the output of the workshop, and the vertical axis shows the price and the cost in taka. The flat line is the price of Tk 40. Each dot shows the marginal cost at the end of one step. The hatched area lies between the price line and the cost line, where each unit adds to profit. The cost line passes above the price line after the fifth step, where each unit would add a loss.

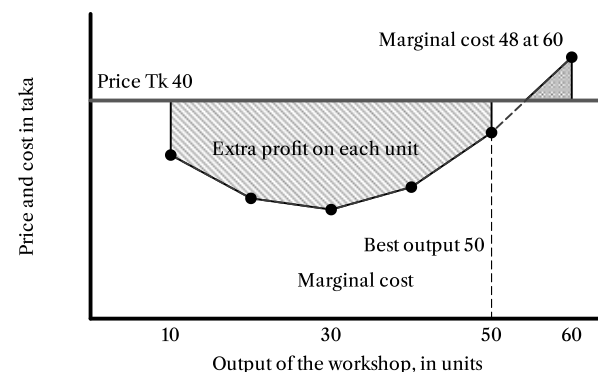


FIG. 13.2. Price taker at marginal cost equal to price: the workshop stops at 50 units

The first five steps all add to profit, and the sixth reduces it by 80. So the workshop makes 50 units. At 50 units, the marginal cost of the last step (34) is below the price (40), and the marginal cost of the next step (48) is above it. The output of 50 is therefore the best

The case file gives the price and the workshop's marginal costs.

The table compares the extra revenue and extra cost of each 10-unit step.

How to read the second chart: the price line, the cost points and the best output.

The workshop adds every step whose extra profit is positive and stops at 50 units.

Before moving on: the nominal wage is in money and the real wage is in goods.

Before moving on. A wage is the payment to labour for its work over a period. The nominal wage is the money amount. The real wage divides it by the price level, so it shows what the wage buys.

* * *

15.3 HOW WAGES ARE SET

WHAT A WORKER ADDS

Return to the shop. The owner is thinking of hiring one more assistant at a wage of Tk 35,000 a month. A sensible owner asks one question: how much more money will the shop receive because of this assistant? If the answer is more than Tk 35,000, hiring pays. If it is less, hiring loses money.

The measure has a name. **Marginal revenue product** means the extra revenue a firm gains from employing one more unit of labour. It combines two things: the extra output the worker produces, and the extra revenue that each unit of output brings, which is the marginal revenue you met earlier. In words: marginal revenue product = extra output from the worker $\times$ marginal revenue per unit. For a worker who produces 20 more books a week, where each extra book brings Tk 500, the marginal revenue product is Tk 10,000 a week.

The rule for the owner follows. Hire one more worker while the marginal revenue product is at least as large as the wage, and stop when it falls below the wage. The extra output from each added worker eventually shrinks, since the workers share the same shop, shelves and counter, and so the marginal revenue product falls as the firm hires more.

Case file, illustrative. At Meghna and Sons, the first extra assistant adds Tk 40,000 a month of revenue, and the second adds Tk 30,000. The wage is Tk 35,000 a month.

Apply the rule. The first assistant brings in Tk 40,000 and costs Tk 35,000, so the shop gains Tk 5,000 a month. The second brings in Tk 30,000 and costs Tk 35,000, so the shop would lose Tk 5,000 a month. The owner hires one extra assistant and stops. Since the figures fall as more are hired, the marginal revenue product schedule works as the firm's demand for labour: it shows how many workers the firm will hire at each wage.

DEMAND AND SUPPLY OF LABOUR

The wage for a whole occupation is set where the demand for labour meets the supply of labour. Demand comes from all the firms that hire; it slopes downward because the marginal revenue product falls as more workers are hired. Supply comes from workers; it slopes upward because a higher wage draws more people into the work, as it draws people from other jobs, from study and from leisure. As in other markets, the equilibrium wage is the wage at which the quantity of labour demanded equals the quantity supplied.

To read the second figure, look at the axes. The vertical axis

A firm hires a worker only if the worker brings in more than the worker costs.

Marginal revenue product. The extra revenue a firm gains from employing one more unit of labour.

Marginal revenue product is the extra revenue from one more unit of labour.

The hiring rule compares the marginal revenue product with the wage.

The record gives the revenue that two extra assistants would add.

The case file shows the rule in use: hire the first assistant but not the second.

A labour market has a demand side and a supply side, like any other market.

To read the second figure, find

Now consider what the owner must earn to continue. **Normal profit** means the least profit a firm must earn to stay in business, covering the owner's implicit costs. Implicit costs are the values of things the owner gives up and does not pay for in money, such as the owner's own time. A baker who could earn a good wage in a factory must earn at least that much from her own bakery, or she will close it.

In this chapter we count normal profit as part of total cost. A profit of Tk 100 is therefore a profit above normal. Chapter 16, 'Interest and Profit', section 16.2 'Profit' gives such a profit its own name and explains where it comes from. Here it is enough to say that a profit above normal attracts new firms, and that a loss drives firms away.

LONG-RUN EQUILIBRIUM

The long run is a period long enough for firms to enter and to leave the industry. Follow the workshop's profit through time in four steps. First, at Tk 40 the workshop earns Tk 100 above normal, and other firms notice it. Second, because entry is free, new firms begin to make notebooks, and the supply of the industry shifts to the right. Third, with more notebooks offered, the market price falls. Fourth, the fall continues until the profit above normal is gone and no new firm has a reason to enter.

The process also runs in reverse. If the price is so low that firms make losses, some firms leave. The supply of the industry shifts to the left, and the market price rises. The rise continues until the remaining firms earn normal profit. In both directions the price moves to the level where no firm earns more than normal profit, and no firm earns less.

The resting point is called *long-run equilibrium*. In it, every firm earns normal profit only, the price equals the lowest cost per unit at which a firm can produce, and no firm wishes to enter or to leave. The consumer gains, because the good sells at the lowest cost at which it can be made. The owner gains nothing above normal, and that is why owners prefer markets where entry is blocked.

The result has limits. It needs free entry, so a blocked market keeps its profit. It assumes that all firms have the same costs, which real firms seldom do. And it takes time: new firms must find money and build workshops, so a profit above normal can last for months. The result gives the direction of the change, and not its speed.

Before moving on. The equilibrium of the firm is the output with the largest profit, where marginal cost equals the price. In the short run a firm may earn a profit above normal or make a loss. In the long run, entry and exit leave every firm with normal profit, at a price equal to the lowest cost per unit.

* * *

WORKED EXAMPLES

The first example uses the workshop's cost data at a lower market

Normal profit is the least payment that keeps the owner in this line of business.

Normal profit. The least profit a firm must earn to stay in business, covering the owner's implicit costs.

In this chapter normal profit counts as a cost, so a profit means a profit above normal.

In the long run, free entry removes profit above normal, and exit removes losses.

Recall: equilibrium price, Chapter 9, section 9.2 'Market Equilibrium'.

The same process runs in reverse when firms make losses.

The resting point is long-run equilibrium: normal profit, and a price equal to the lowest cost per unit.

The result holds under conditions: free entry, equal costs and enough time.

Before moving on, recall the short run, the long run and normal profit.

Worked example 1 finds the

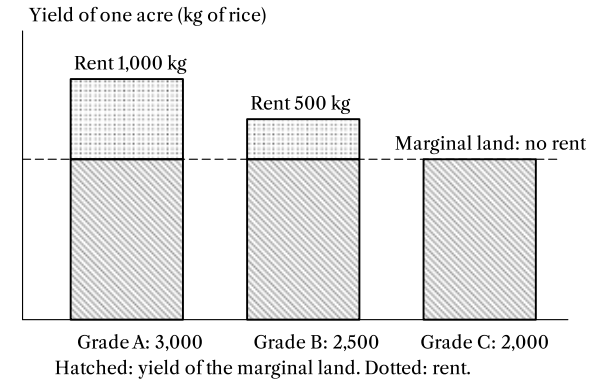


FIG. 15.1. Differential rent as stacked grades of land

Ricardo's theory applies to shop sites as well as to farms, and the next record shows it.

Case file, illustrative. Meghna and Sons trades at a site in Dhanmondi, Dhaka. The shop's profit before rent at Dhanmondi is Tk 1,470,000 a year. At the marginal site, which pays no rent, the profit before rent would be Tk 750,000. The difference is Tk 720,000, which equals the rent paid: Tk 60,000 $\times$ 12 = Tk 720,000.

The record shows the theory at work. A tenant would not pay more than Tk 720,000 a year for the Dhanmondi site, because the marginal site yields Tk 750,000 for no rent. The rent also cannot fall far below Tk 720,000, because other shopkeepers would bid for such a profitable site. Competition between users therefore pushes the rent to the difference. Note that rent does not cause the price of the goods sold at the site. It is the result of the site's advantage.

PAUSE AND CHECK 15.P1 Three grades of land yield 90, 70 and 60 bags an acre, and the 60-bag land is the marginal land. Find the rent of each grade, in bags.

Before moving on. In Ricardo's theory, land of different quality earns differential rent, equal to the difference between its yield and that of the marginal land. The marginal land pays no rent. The same reasoning applies to shop sites.

QUASI-RENT

Land is a gift of nature. Is there a similar surplus for things people make? Yes, but it does not last. The English economist Alfred Marshall, in his *Principles of Economics* (1890), gave it a name.

Quasi-rent means a temporary surplus earned by a man-made asset while its supply is fixed in the short run. Suppose the demand for printed books rises. The number of printing presses cannot rise overnight, so those that exist earn more than the least needed to keep them running, namely their variable costs. The surplus above variable cost is what pays the fixed cost and leaves a gain. In the long run, new presses are built, supply rises and the

Ricardo's theory applies to shop sites as well as farms.

The record gives two profits and the rent they imply.

The case file confirms that rent equals the gain of the better site over the marginal one.

Before moving on: Ricardian rent is the gain of a better plot over the marginal plot.

Man-made assets can earn a surplus for a short time.

Quasi-rent. A temporary surplus earned by a man-made asset while its supply is fixed in the short run.

Quasi-rent is a short-run surplus earned by a man-made asset.

13.T1 A firm in perfect competition earns a profit above normal this year. Explain why this profit is unlikely to last, using free entry and the shift of the industry's supply.

13.T2 The potato seller must accept the market price, but the only pharmacist for ten kilometres can raise a price and still keep customers. Name two features of their markets that explain the difference.

WHERE PEOPLE GO WRONG

Mistake: "To earn the most, the workshop should make as many units as it can." The sixth step of the workshop adds Tk 480 to cost and only Tk 400 to revenue, so it reduces profit. The corrected rule: raise output only while each step adds to profit, and stop where marginal cost reaches the price.

Mistake: "A price taker is any firm that is small." A price taker is a firm whose sales are too small *relative to the market* to affect the price. A very large firm in a very large market can be a price taker, while a small firm that is the only seller in a village is not. The corrected rule: judge the firm against its market.

Mistake: "In the long run the owner earns nothing." The owner earns normal profit, which pays for the owner's time and money. What vanishes is the profit above normal. The corrected rule: zero profit above normal means that the owner is paid exactly as much as the next best use of the owner's effort.

IN HISTORY

In 1848, a group of businessmen in Chicago, in the United States, founded the Chicago Board of Trade. Their aim was to bring order to the grain market of the American Midwest, which had been disorderly. The exchange gave dealers one place to meet, and its trading produced prices that every member could see.² For much of its history, traders there called out their bids and offers face to face in trading rooms known as pits.

Economists use a grain exchange as an example because it comes close to the model of this chapter: many dealers traded the same kind of grain, and each price was called for all to hear. The match was not perfect. Trading on the floor was open to members, so entry was not entirely free, and no real market meets all four conditions.

REVIEW QUESTIONS

13.1 A firm in perfect competition sells its good at Tk 10. Its marginal cost for each 5-unit step is 4, 7, 9 and 12, at outputs 5 to 20. Find the best output and the extra profit of the steps that it makes.

13.2 The market price of a notebook rises to Tk 50. Using the binding workshop's marginal costs (30, 22, 20, 24, 34 and 48 for each

² The exchange later added futures contracts and other products. Those later developments lie outside this chapter.

Mistake one: believing that the largest output gives the largest profit.

Mistake two: believing that a price taker is always a small firm in a small town.

Mistake three: believing that zero profit in the long run means the owner earns nothing.

In 1848 grain dealers in Chicago founded an exchange to bring order to a chaotic market.

Such an exchange comes close to perfect competition but meets not every condition.

Five questions follow the ladder from a clean example to a combined problem.

YOU WILL NEED

Chapter 14, 'Monopoly and Imperfect Competition', section 14.1 'Monopoly', for monopoly; Chapter 9, 'Elasticity of Supply and Market Equilibrium', section 9.3 'Changes in Equilibrium', for changes in equilibrium; Chapter 11, 'Costs of Production', section 11.1 'Explicit and Implicit Costs', for explicit and implicit costs; and Appendix A1, 'Maths You Will Use', for percentages and index numbers.

A hiring fair brings together owners of land and sellers of labour.

Two puzzles stand beside each other: the payment to land and the payment to workers.

The chapter answers both puzzles by looking at demand and supply.

On a market day in an old English county, farmers drive their carts into the square and workers stand beside them, waiting to be hired for the year. A farmer with two fields looks over the crowd. Everyone at the fair knows that the farmer will pay the labourers, for they have to be tempted to work. Few ask a stranger question: why does the farmer's field earn a payment at all? Nobody had to be tempted to make the field.

A second puzzle sits close to the first. In the crowd, a ploughman asks for less than a skilled stonemason, and the stonemason asks for less than a surgeon would in the town. They are all people. Why do their payments differ so widely? And what stops a farmer from offering half the going rate, or a worker from demanding double?

The answers come from the tools you already have. Section 15.1 studies rent, the payment to land, and shows that a whole class of payments share its character. Section 15.2 turns to wages and distinguishes the wage in money from the wage in goods. Section 15.3 uses demand and supply to explain how wages are set and why they differ.

* * *

15.1 RENT

RENT AND ECONOMIC RENT

Begin with a familiar payment. A family pays a landlord each month to live in a flat. A farmer pays an owner each year to plough a field. In both cases a person pays for the use of land that belongs to someone else. This payment is **rent**, which means the payment to the owner of land for its use, in everyday talk and in economics. A tenant who pays Tk 60,000 a month for a shop site is paying rent. In everyday speech the word is also used for cars and machines, so this book keeps it for land unless it says otherwise.

Land has a feature that most goods lack. Nobody can make more of it. The whole supply of land in a country is fixed, whatever the payment. In words you know from Chapter 9, the elasticity of supply of land as a whole is zero. Because supply cannot respond, the payment to land is set by demand alone: when many people



CHAPTER XIV

Monopoly and Imperfect Competition

Who sets the price when there is no crowd of sellers?



Chapter 13 studied a firm so small that it must accept the market price. This chapter studies the other cases: the single seller, the many sellers whose goods differ, and the few large sellers who watch one another.

14.4 Suppose a competitive market faced the demand and marginal cost of the case file, so that price equals marginal cost. Find its output and price, and compare them with the monopoly's.

14.5 Name the market: (a) many tailors, each with a different style, free to open new shops; (b) three firms supply all the cement in a country; (c) one firm holds the only licence to supply a town with water.

14.6 A tea stall's total revenue is Tk 800 at 40 cups and Tk 900 at 50 cups, and its marginal cost is Tk 12 a cup for those 10 cups. Find the marginal revenue per cup and say whether the stall should sell the extra 10 cups. Is marginal revenue below the price of Tk 18 at 50 cups?

price.¹ The monopolist's power has a limit. If it raises its price, consumers buy less; if it lowers its price, they buy more. It can choose the price or the quantity, but not both: the demand curve fixes the other.

A monopoly lasts only while rivals cannot enter. Four common barriers stand in their way. The firm may own the only supply of a key resource, as one mine may hold all the stones. The law may give it an exclusive right, such as a patent or a licence. One large firm may serve the market at a lower cost for each unit than two smaller firms could, so a second firm would lose money. Or the firm may use threats to drive rivals away. The law differs by country: some governments grant exclusive rights, and others forbid firms to misuse their power, so a reader must learn the rules of the country concerned.

Return to the bookshop. To choose her output, the owner must know what each extra copy adds to her revenue. This idea is called **marginal revenue**, which means the extra revenue from selling one more unit of output. If she sells one more copy and her revenue rises by Tk 40, the marginal revenue is Tk 40. For a price taker, marginal revenue equals the price, because extra units sell at the same price. For a price maker it is smaller than the price, and the next paragraph shows why.

Take the case that follows. The seller sells 20 copies at Tk 60. To sell 30 copies she must cut the price to Tk 50, and the cut applies to all 30 copies, including the first 20. She gains 10 new copies at Tk 50, which is Tk 500. She loses Tk 10 on each of the first 20 copies, which is Tk 200. The net gain is Tk 300 for 10 extra copies, or Tk 30 a copy. So marginal revenue, Tk 30, lies well below the price of Tk 50.

Given. At 20 units the price is Tk 60 and total revenue is Tk 1,200; at 30 units the price is Tk 50 and total revenue is Tk 1,500. *Find.* The marginal revenue per unit for the step from 20 to 30 units. *Formula.* In words: the rise in total revenue divided by the rise in output. In symbols: $MR = X \div Y$, where MR is marginal revenue, X is the rise in total revenue and Y is the rise in output. *Steps.* $X = 1,500 - 1,200 = \text{Tk } 300$. $Y = 30 - 20 = 10$ units. $MR = 300 \div 10 = \text{Tk } 30$ per unit. *Answer.* Marginal revenue for this step is Tk 30 per unit. *Check.* The gain of Tk 500 less the loss of Tk 200 is Tk 300, which agrees; and Tk 30 is below the price of Tk 50, as the argument said it must be.

PAUSE AND CHECK 14.P1 A seller sells 20 units at Tk 60 and 30 units at Tk 50. Why is the marginal revenue of the step below Tk 50?

Case file, illustrative. A seller with the whole market for a specialised title: price Tk 70 at 10 units, Tk 60 at 20, Tk 50 at 30, Tk 40 at 40, Tk 30 at 50 (price = $80 - \text{quantity}$); total revenue 700, 1,200, 1,500, 1,600, 1,500; marginal revenue per step 70, 50, 30, 10, -10; marginal cost per step 30, 22, 20, 24, 34. Output 30 at price Tk 50; total cost 1,320; profit Tk 180.

The table repeats the case file in a form that is easy to scan. Read

¹ Recall: price taker, Chapter 13, 'Market Structure and Perfect Competition', section 13.2 'The Perfectly Competitive Firm'.

Barriers to entry protect a monopoly from rivals.

Marginal revenue. The extra revenue from selling one more unit of output.

To choose her output, the seller needs the extra revenue from one more unit.

To sell more, a monopolist must cut the price on every unit.

A worked example turns the idea into a formula.

The case file gives the seller's price, revenue and costs.

The table sets the case out in five rows.

has over price. *You should now be able to say* which market a given description fits.

WORKED CASE: THE SELLER'S BEST OUTPUT

Given. From the case file of section 14.1: demand price = $80 - \text{quantity}$; total revenue 700, 1,200, 1,500, 1,600, 1,500 at outputs 10 to 50; marginal cost per unit for each 10-unit step 30, 22, 20, 24, 34; total cost Tk 1,320 at 30 units. *Find.* The output of largest profit, the price, and the profit. *Formula.* In words: raise output while marginal revenue is at least marginal cost; read the price from demand; profit is total revenue minus total cost. In symbols: profit = $P \times Q - C$, where P is price, Q is output and C is total cost.

Steps. (1) Marginal revenue per unit for each step: $700 \div 10 = 70$; $(1,200 - 700) \div 10 = 50$; $(1,500 - 1,200) \div 10 = 30$; $(1,600 - 1,500) \div 10 = 10$; $(1,500 - 1,600) \div 10 = -10$. (2) Compare with marginal cost: 70 against 30, take the step; 50 against 22, take it; 30 against 20, take it; 10 against 24, stop. (3) Output is 30 units. (4) Price = $80 - 30 = \text{Tk } 50$. (5) Total revenue = $50 \times 30 = \text{Tk } 1,500$. (6) Profit = $1,500 - 1,320 = \text{Tk } 180$. *Answer.* The seller sells 30 units at Tk 50 and earns a profit of Tk 180. *Check.* Total cost at 20 units is $1,320 - 10 \times 20 = \text{Tk } 1,120$, so profit there is $1,200 - 1,120 = \text{Tk } 80$. Total cost at 40 units is $1,320 + 10 \times 24 = \text{Tk } 1,560$, so profit there is $1,600 - 1,560 = \text{Tk } 40$. Both are below Tk 180.

Given. Price = $60 - \text{quantity}$; fixed cost Tk 100; marginal cost per unit for the 10-unit steps ending at 10, 20, 30 and 40 units: 20, 22, 25, 30. *Find.* Output, price and profit. *Formula.* As above. *Steps.* Total revenue at 10, 20, 30, 40 units: 500, 800, 900, 800. Marginal revenue per unit: 50, 30, 10, -10. Against marginal cost, the first two steps are taken (50 over 20, 30 over 22) and the third is not (10 under 25). Output is 20 units and price is $60 - 20 = \text{Tk } 40$. Total cost = $100 + 10 \times 20 + 10 \times 22 = \text{Tk } 520$. Profit = $800 - 520 = \text{Tk } 280$. *Answer.* 20 units at Tk 40, profit Tk 280. *Check.* At 30 units, total cost is $520 + 10 \times 25 = 770$ and revenue is 900, so profit is 130, which is less than 280.

Common slips deserve a warning. Some students compare price with marginal cost instead of marginal revenue with marginal cost. Some read the price from the marginal revenue line, when it must come from the demand curve at the chosen output. Some forget that a step of 10 units must be divided by 10 to give a figure per unit. Others choose an output where marginal revenue is negative, which always loses money when marginal cost is positive.

Before moving on. To find a monopolist's best output, compute marginal revenue for each step, take steps while it is at least marginal cost, read the price from demand, and then find profit. *You should now be able to say* how each of the six steps leads to the answer.

* * *

SUMMARY

usually higher and the output lower than in a competitive market, as section 14.4 shows.

Before moving on. A monopoly has one seller and no close substitute, so the seller is a price maker. Marginal revenue lies below price, and the seller chooses the output at which marginal revenue meets marginal cost, then reads the price from the demand curve. *You should now be able to say* why a monopolist's marginal revenue is smaller than its price.

14.2 MONOPOLISTIC COMPETITION

Walk down a busy street in Dhaka lined with small stalls that sell tea and biscuits. There are many sellers, and anyone may open another stall. Yet no two stalls are the same. One has a shaded seat, another brews a stronger tea, a third stands where the buses stop. Regular consumers prefer one stall and will pay a little more to keep going there. Each stall therefore holds a small power over its own price, although many rivals are close by.

This market has a name. **Monopolistic competition** is a market with many sellers of similar but not identical goods, and free entry. The tea street is one example; hair salons and small restaurants in one neighbourhood are others. The name joins two ideas. Each seller is a little like a monopolist, because its good differs from rivals' goods. Each seller also competes, because rivals offer close, though not perfect, substitutes, and anyone may enter.

The source of each seller's small power is **product differentiation**, which means making a good different from rivals' goods in quality, style or brand, to build loyal buyers. A stall that serves tea in clean glasses with a friendly greeting has differentiated itself. The *brand* of a good is the name and style by which consumers recognise it. The differences may be real, such as a stronger tea, or only believed, such as a famous name. Either way, a loyal consumer finds the good hard to replace.

Because the good is different, a seller can raise its price without losing every consumer, so its demand curve slopes downward, as a monopolist's does. Because close substitutes exist, a rise in its price still sends many consumers to rivals, so the curve is flatter than a monopolist's would be. Like the monopolist, the seller finds that marginal revenue lies below price, and it chooses the output at which marginal revenue equals marginal cost. We call that choice the equilibrium of the firm, a term taught in Chapter 13, section 13.3 'Equilibrium of Firm and Industry'.

In the short run, a differentiated seller may earn more than the least profit it needs to stay in business. That profit attracts new sellers, who open new stalls with similar goods. Each new rival takes some consumers from the existing sellers, so every existing seller's demand curve shifts inwards. The process continues until a typical seller earns only enough to stay in business. This is the same

Recap of section 14.1.

A Street of tea stalls shows many sellers whose goods differ.

Monopolistic competition.

A market with many sellers of similar but not identical goods, and free entry.

Many sellers of similar but different goods, with free entry, make monopolistic competition.

Product differentiation.

Making a good different from rivals' goods in quality, style or brand, to build loyal buyers.

Brand. Word in use. In daily life, a name printed on a good, or a mark burned on cattle. In economics, a name and style that make a firm's good known and different from rivals' goods.

Differences in quality, style or brand give each seller loyal consumers.

Its demand curve slopes down but is flatter than a monopolist's.

Free entry removes extra profit over time.

Recap of section 14.2.

Three bus firms on one road show how few sellers watch each other.

Oligopoly. A market dominated by a few large sellers, each aware of what the others do.

A few large sellers, each watching the others, make an oligopoly.

Each firm's best choice depends on its rivals' choices.

Kinked demand explains why prices in an oligopoly often stay fixed.

Firms may compete on features or agree to act together, and the law often forbids agreement.

From the consumer's viewpoint, it is choice. From the employee's viewpoint, a stall with loyal consumers may offer steadier work.

Before moving on. Monopolistic competition has many sellers, free entry and goods that differ. Product differentiation gives each seller a downward-sloping demand curve, flatter than a monopolist's, and free entry removes extra profit over time. *You should now be able to say* what separates it from both perfect competition and monopoly.

14.3 OLIGOPOLY

Imagine a long road served by only three bus firms. If one firm cuts its fare, it will take many riders from the other two, so they will probably cut theirs too. If one raises its fare, the others may keep theirs low and take its riders. Before changing a fare, each firm must guess how the others will react. A seller among many sellers never worries about this. A seller among a few worries about it all the time.

An **oligopoly** is a market dominated by a few large sellers, each aware of what the others do. Mobile telephone service, flights on a busy route and cement in a country are common examples. The word comes from Greek words meaning *few* and *to sell*. Entry is hard, because the firms are large and a newcomer needs a great deal of money, equipment or consumer loyalty to compete with them.

The key feature is *interdependence*: the best price or output of each firm depends on what its rivals do. In a competitive market a firm ignores its rivals, because each is too small to matter. In a monopoly there are no rivals to watch. In an oligopoly the firms are few enough that every move is noticed. This makes the market harder to predict, and economists offer several theories rather than one.

One theory explains why oligopoly prices often stay the same for long periods. Suppose a firm believes that its rivals will copy a price cut but ignore a price rise. If it cuts its price, rivals match the cut, and it gains few extra consumers. If it raises its price, rivals do not follow, and it loses many consumers. So its demand curve bends, or has a kink, at the present price: flat above that price and steep below it. The firm sees little reward in changing its price in either direction, so it leaves the price alone. The theory explains why a price stays where it is; it does not explain how the price was first set.

PAUSE AND CHECK 14.P2 Why might an oligopolist leave its price unchanged even when its costs fall a little?

Oligopolists have two broad paths. They may compete without cutting prices, by advertising, quality and service. Or they may agree with one another on prices or output, acting together like a single monopolist. Many countries forbid such agreements through competition law, but the rules and the penalties differ from country to country. Agreements are also fragile, because each firm can gain by secretly selling below the agreed price, and when many

Why is the rate 11 per cent and not 5 or 20? One answer treats money as a good that is bought and sold. **Loanable funds** are the money that savers offer and borrowers demand at each interest rate. Savers are people and firms with money they do not need now. Borrowers are people and firms that want money to spend now. The loanable funds theory says that the rate of interest is the price at which the funds offered equal the funds demanded.

The first figure below shows the idea. To read it, look at the horizontal axis, which measures loanable funds in taka, and the vertical axis, which measures the rate of interest in per cent a year. The falling line shows that borrowers want more funds when the rate is low, because borrowing is cheap. The rising line shows that savers offer more funds when the rate is high, because waiting is better rewarded. The lines cross at one rate. At that rate, the funds that borrowers demand equal the funds that savers offer.

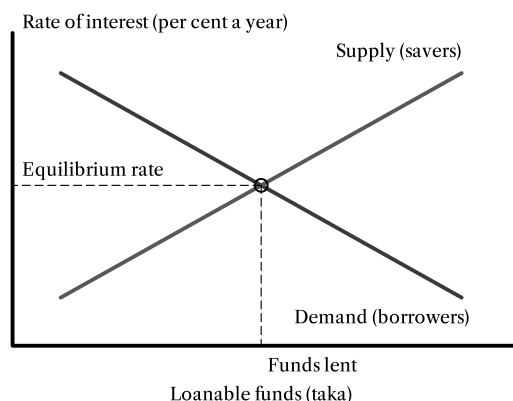


FIG. 16.1. Demand for and supply of loanable funds, and the rate of interest where they meet

The theory predicts how the rate moves. If people begin to save more, the rising line moves to the right, and the rate falls. If firms want to borrow more to build factories, the falling line moves to the right, and the rate rises. Many students find it odd that a rate can be a price. The oddity disappears when you remember that the thing being bought is the use of money for a year.

A second theory looks at the other side of the choice. **Liquidity preference** is the wish of people to hold money in cash form rather than lend it. Cash can be spent at once, which is useful. John Maynard Keynes, a British economist, argued in 1936 that people hold cash to pay for daily purchases, to be ready for emergencies, and to wait for better chances to buy assets. A lender who gives up cash loses these benefits. Interest is the reward that persuades people to part with cash.

Both theories have limits. The loanable funds theory stresses savers and borrowers. The liquidity preference theory stresses the supply of money and the wish to hold it. A country's central bank, such as Bangladesh Bank, also influences rates through its policy. In

The rate of interest is set where the wishes of savers and borrowers meet.

Loanable funds. The money that savers offer and borrowers demand at each interest rate.

Borrowers want more at a low rate, and savers offer more at a high rate.

If savers offer more, the rate falls, and if borrowers want more, the rate rises.

A second theory says that people hold money because holding it is useful.

Liquidity preference. The wish of people to hold money in cash form rather than lend it.

Both theories explain interest, and neither is complete.

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THE THREE RESULTS OF ECONOMIC PROFIT

ECONOMIC PROFIT	WHAT IT MEANS	LIKELY RESULT
Above zero	Supernormal profit: more than the owner could earn elsewhere	New firms are attracted
Equal to zero	Normal profit only: the owner earns exactly what is needed	The owner stays
Below zero	Less than the owner could earn elsewhere	The owner may leave

Read the table one row at a time. In the first row, the owner earns more than the best alternative, so other people want to enter the same line of business. In the second row, the owner is neither tempted nor driven away. In the third row, the owner would do better to use time and money elsewhere, although some owners stay for reasons that money does not measure.

The measure matters because owners compare their own results with the best alternative. When economic profit is above zero in an industry, other people notice and open similar firms. When it is below zero for long, owners move their time and money to other uses. Accounting profit cannot give this guidance, because it leaves out the alternatives. A limit: implicit costs are estimates, so economic profit is an estimate too.

Case file, illustrative. For 2025, Meghna and Sons recorded accounting profit of Tk 750,000. The case file counts two implicit costs: Tk 600,000 for the owner's time, and Tk 287,000 for the interest forgone on capital employed of Tk 4,100,000 at 7 per cent. Together, the implicit costs are Tk 887,000. Economic profit is therefore 750,000 minus 887,000, which is -Tk 137,000.

The shop shows the difference between the two measures. The accounts report a gain of Tk 750,000, and any lender or tax office would see a firm with a positive profit. Yet the owner's own resources could have earned Tk 887,000 elsewhere. From the owner's viewpoint, the shop earns Tk 137,000 less than the best alternative. The employee and the customer see nothing of this. Only the owner pays the implicit costs.

PAUSE AND CHECK 16.P2 A firm has revenue of Tk 900,000 and explicit costs of Tk 700,000. Its owner's time and capital could earn Tk 150,000 elsewhere. Find the accounting profit and the economic profit.

Before moving on. Accounting profit subtracts explicit costs only. Economic profit subtracts explicit and implicit costs, so it also subtracts normal profit. Economic profit above zero is supernormal profit. You should now be able to say why a firm can have a positive accounting profit and a negative economic profit.

* * *

16.3 WHY PROFIT OCCURS

Interest has a clear reason: lenders wait. Profit is harder to explain

Each row of the table pairs one result with its likely consequence.

Economic profit guides decisions to enter, stay in or leave a line of business.

The shop's records contain all the figures needed for both measures.

The shop has a positive accounting profit and a negative economic profit.

The section ends with a recap.

If profit can be negative, we

others. You should now be able to name the four views and give one limit of each.

* * *

WORKED EXAMPLE: INTEREST AND ECONOMIC PROFIT OF THE SHOP

We work two examples in the same order each time: Given, Find, Formula, Steps, Answer and Check. The first finds the interest on the shop's bank loan. The second finds the shop's economic profit.

Given. The shop owes Tk 600,000 to its bank. The rate of interest is 11 per cent a year, and the loan runs for one year. *Find.* The interest for the year. *Formula.* In words, interest equals the sum borrowed multiplied by the rate. In symbols, $I = S \times r$, where I is the interest, S is the sum borrowed and r is the rate written as a decimal.

Steps. Write 11 per cent as $11 \div 100 = 0.11$. Then $I = 600,000 \times 0.11 = \text{Tk } 66,000$. *Answer.* The shop pays Tk 66,000 in interest for the year. *Check.* One per cent of 600,000 is Tk 6,000. Eleven per cent is eleven times that, which is $11 \times 6,000 = \text{Tk } 66,000$. The two routes agree.

Given. Sales were Tk 14,400,000, and accounting profit was Tk 750,000. The owner's time is valued at Tk 600,000. Capital employed was Tk 4,100,000, and it could have earned 7 per cent a year elsewhere. *Find.* The economic profit. *Formula.* Forgone interest = rate $\times$ capital employed. Economic profit = accounting profit - total implicit costs. This is the same as revenue minus all costs, because accounting profit is revenue minus explicit costs.

Steps. Write 7 per cent as 0.07. Forgone interest = $0.07 \times 4,100,000 = \text{Tk } 287,000$. Total implicit costs = $600,000 + 287,000 = \text{Tk } 887,000$. Economic profit = $750,000 - 887,000 = -\text{Tk } 137,000$.

Answer. The economic profit is -Tk 137,000. The shop earns Tk 137,000 less than its owner's time and capital could earn elsewhere. *Check.* The implicit costs of Tk 887,000 exceed the accounting profit of Tk 750,000, so the answer must be negative. Adding back: $-137,000 + 887,000 = 750,000$, the figure in the accounts. A negative result does not force the owner to close the shop. It tells her the price of staying.

Common slips are easy to name. First, using 11 instead of 0.11, which gives an answer one hundred times too large. Second, mixing periods, such as a yearly rate with a loan held for three months. Third, forgetting implicit costs, which gives accounting profit and not economic profit. Fourth, a wrong sign, such as subtracting the smaller number from the larger. Write the unit beside every number, and check the sign by asking whether the implicit costs exceed the accounting profit.

* * *

SUMMARY

280,000. Economic profit = $300,000 - 280,000 = \text{Tk } 20,000$, which is supernormal profit.

16.7 Risk bearing, limit: it does not pay every risk-taker. Marginal productivity, limit: one owner's contribution is hard to measure. Innovation, limit: many profits need no new idea. Monopoly, limit: it does not explain profit among many rivals.

Two examples apply the ideas, one small and clean and one realistic.

The first example finds a year's interest on the bank loan.

Each line of arithmetic is shown, with its unit, so the method is the only new thing.

The second example finds the economic profit of the shop for 2025.

Every step is written out, so nothing depends on guesswork.

The shop is below normal profit by Tk 137,000.

Four common slips spoil these calculations.

16.2 A loan of Tk 450,000 is made at 9.5 per cent a year for one year. Find the interest.

16.3 A loan of Tk 600,000 at 11 per cent a year is held for three months. Interest is charged on the sum borrowed only and is not added to it. Find the interest.

16.4 An owner's accounting profit is Tk 420,000. Her time is worth Tk 300,000 a year elsewhere, and her capital of Tk 2,000,000 could earn 6 per cent a year. Find the economic profit and say which of the three results in the table it is.

16.5 A firm has an accounting profit of Tk 380,000 and implicit costs of Tk 450,000. Find the economic profit and say what it tells the owner.

16.6 A trader has sales of Tk 2,000,000 and explicit costs of Tk 1,700,000. Her time is worth Tk 200,000 a year elsewhere, and her capital of Tk 1,000,000 could earn 8 per cent. Find her accounting profit and economic profit, and name the result.

16.7 Name the four views of why profit occurs, and give one limit of each.

seller for ten kilometres, so customers have little choice and entry of rivals is unlikely.

13.1 Extra revenue per step is $5 \times 10 = 50$. Extra costs are 20, 35, 45, 60, so extra profits are +30, +15, +5, -10. The best output is 15 units, and the steps made add $30 + 15 + 5 = \text{Tk } 50$.

13.2 Extra revenue per step is 500; extra costs are 300, 220, 200, 240, 340, 480, so every step adds profit and the best output is 60 units. Total revenue is $60 \times 50 = 3,000$, total cost is $1,900 + 480 = 2,380$, and profit is Tk 620.

13.3 Each step earns extra revenue of $100 \times 20 = 2,000$ against extra costs of 1,800, 1,400, 1,600, 2,200, 3,000. Extra profits are +200, +600, +400, -200, -1,000. The best output is 300 kilograms, and the steps made add Tk 1,200.

13.4 New demand: $100 - \text{price} = \text{price}$, so the price is Tk 50. The workshop's best output is 60 units and its profit is Tk 620, against Tk 100 at Tk 40. Profit above normal attracts new firms, so industry supply shifts right and the price falls in the long run.

13.5 Monopolistic competition: many sellers, different versions of the good (each tailor has a style) and easy entry.

CHAPTER XIV · MONOPOLY AND IMPERFECT COMPETITION

14.p1 To sell 30 units she must cut the price on all units. She loses Tk 10 on each of the first 20 units (Tk 200) and gains 10 new units at Tk 50 (Tk 500). The net gain is Tk 300, or Tk 30 a unit, which is below Tk 50.

14.p2 Each firm fears that rivals would copy a price cut but not a price rise, so a change brings it little gain either way. It leaves the price alone.

14.t1 No. One large firm may serve everyone at a lower cost for each unit than many small firms, and a market of differentiated goods gives variety that identical goods cannot. Price is only one measure of how well a market serves consumers.

14.t2 At 50 units marginal revenue is -10, so the extra units lower revenue while adding cost. Output of 40 units gives revenue Tk 1,600 against Tk 1,500 at 50, and cost falls from Tk 1,900 to Tk 1,560, so profit rises from -Tk 400 to Tk 40.

14.1 $(1,600 - 1,500) \div (40 - 30) = 100 \div 10 = \text{Tk } 10$ per unit.

14.2 Marginal revenue per unit: 50, 30, 10, -10. Take the first two steps (50 over 20, 30 over 22); stop at the third (10 under 25). Output 20 units, price $60 - 20 = \text{Tk } 40$. Total cost = $100 + 200 + 220 = \text{Tk } 520$. Profit = $800 - 520 = \text{Tk } 280$.

14.3 Tk 300 for the whole step of 10 units is $\text{Tk } 300 \div 10 = \text{Tk } 30$ per unit, so the figures agree. Compare the per-unit figure, Tk 30, with the marginal cost of Tk 20 per unit; marginal revenue is higher, so the step is worth taking.

14.4 At 40 units the price is Tk 40 and the marginal cost is Tk 24, so output rises; at 50 units the price is Tk 30 and the marginal cost is Tk 34, so it stops. Competitive output is 40 units at Tk 40, against 30 units at Tk 50 for the monopoly: ten more units at Tk 10 less.

GLOSSARY

Accounting profit. Revenue minus explicit costs only. Revenue is the money a firm receives from sales. Accounting profit appears in a firm's accounts. A farmer who sells a harvest for Tk 300,000 and pays Tk 250,000 in costs has an accounting profit of Tk 50,000. (*Chapter XVI*)

Average product. Total product divided by the number of units of the input used. It shows the output per unit. Example: $44 \div 4 = 11$ with 4 workers. (*Chapter X*)

Average total cost. Total cost divided by the number of units produced, so that fixed and variable costs are shared among all units. It is the cost of each unit on average. A total cost of Tk 1,120 for 20 books gives Tk 56 a book. (*Chapter XI*)

Bandwagon effect. A rise in demand for a good because many other people buy it. A buyer wants what friends and neighbours have, as when a mobile phone becomes more useful because all of one's friends use it. (*Chapter IV*)

Budget line. A line on a graph showing every combination of two goods that a consumer can buy with a given income at given prices. Its ends are income divided by each price, and its steepness is the price ratio. Example: Tk 400 at Tk 40 and Tk 20 gives ends of 10 notebooks and 20 pens. (*Chapter VII*)

Ceteris paribus. A Latin phrase meaning all other things stay the same, used to study one change at a time. It is a method of thinking, not a claim that nothing else changes. To study the effect of price on notebook sales, we hold incomes and other prices fixed. (*Chapter I*)

Competitive market. A market with many buyers and sellers, none of whom can set the price. Each one must accept the price that the whole market produces. A street of tea stalls selling the same tea is a small example. (*Chapter IX*)

Complementary goods. Goods used together, so a price rise for one lowers demand for the other. Pens and notebooks are used together. If pens become dearer, fewer notebooks are wanted at every price. (*Chapter IV*)

Consumer equilibrium. The point where a consumer gets the greatest satisfaction possible from a given income. It lies on the budget line, on the highest reachable indifference curve, where the marginal rate of substitution equals the price ratio. Example: 6 notebooks and 8 pens. (*Chapter VII*)

Consumers' surplus. The gap between what buyers are willing to pay for a good and what they actually pay. Added over all units bought, it measures the gain consumers keep. In the notebook case, a price of Tk 40 leaves three customers a surplus of Tk 30 in total. On a chart it is the triangle under the demand curve and above the price. (*Chapter VI*)

Consumption. The use of goods and services by people to satisfy their wants. Eating a meal, riding a bus and filling a notebook with notes are all consumption. It takes place when the good is used

9.5 $90 - \text{price} = 2 \times \text{price} - 30$, so $120 = 3 \times \text{price}$, $\text{price} = 40$, quantity = 50. Compared with the demand shift only (price 30, quantity 60), the price is higher and the quantity is lower.

CHAPTER X · PRODUCTION FUNCTION AND RETURNS

10.p1 Marginal product of the third helper = $18 - 12 = 6$ loaves. Average product with 3 helpers = $18 \div 3 = 6$ loaves.

10.p2 The 8 means the fourth worker adds 8 to total product ($44 - 36$). Average product with 4 workers is $44 \div 4 = 11$.

10.p3 With some inputs fixed, extra units of one input eventually add less and less to output. The fixed input is the machine.

10.t1 The machine is already used to its limit and the bench is crowded, so the extra worker has no machine time to use. The cause is the fixed input, not the skill of the worker.

10.t2 Each extra worker adds less output for the same wage, so more worker-days, and more pay, are needed for each extra unit of output.

10.1 A production function is a rule showing the largest output that can be made from each combination of inputs. It records the best method known, not what a careless firm might produce.

10.2 In the short run at least one input is fixed; in the long run all inputs can be changed. A machine or an oven is a fixed input.

10.3 Marginal product: 30, 40, 30, 20. Average product: 30, 35, 33.33, 30.

10.4 The third worker's marginal product is 12, equal to the average of 12 for 2 workers, so the average does not change.

10.5 Output rises by less than the inputs (2.4 against 3): decreasing returns to scale.

10.6 Output rises by more than the inputs (2.6 against 2): increasing returns to scale.

10.7 The law changes one input and holds others fixed, and it belongs to the short run; returns to scale changes all inputs together, and it belongs to the long run. The law concerns the extra output; returns to scale compares proportional changes.

CHAPTER XI · COSTS OF PRODUCTION

11.p1 Yes. It is an implicit cost, because the building has a best alternative use worth Tk 5,000 a month, although she pays nothing in money.

11.p2 Total cost rises by $1,560 - 1,320 = \text{Tk } 240$ for 10 books, so marginal cost is $240 \div 10 = \text{Tk } 24$ a book.

11.t1 Average variable cost rises by Tk 2, but average fixed cost falls from Tk 15 to Tk 12, a fall of Tk 3. The net change is -Tk 1. Marginal cost of Tk 34 is also below the old average of Tk 39, which pulls the average down.

11.t2 Her time is an implicit cost, because it could earn a wage elsewhere. She should ask: what is the best use of my time outside the shop, and what is it worth?

11.1 $\text{TC} = \text{FC} + \text{VC} = 2,000 + 3,000 = \text{Tk } 5,000$. $\text{MC} = \text{VC}' = 10Q$. $\text{MC} = \text{TC}' = 10Q$.

Excess supply. A situation in which sellers offer more than buyers want at a price. Some goods stay unsold, so sellers cut the price and push it down. In the notebook market at Tk 50, firms offer 50 and buyers want 30. (*Chapter IX*)

Explicit cost. A cost that involves a payment of money to people outside the firm, such as rent, wages, paper or interest on a loan. It appears in a receipt or bank record. A shop's monthly rent paid to its landlord is an example. (*Chapter XI*)

Fixed cost. A cost that does not change with the quantity produced in the short run, because it is owed even when nothing is made. It is the cost of being ready to produce. A workshop's machine rent is an example. (*Chapter XI*)

Giffen good. A rare, very basic good for which quantity demanded rises when its price rises. Marshall wrote that Giffen noticed it among very poor households buying bread. It is a special case and the law of demand holds for nearly all other goods. (*Chapter IV*)

Implicit cost. The value of resources that the owner supplies to the firm, for which no money is paid. These resources could have earned money elsewhere, so they have an opportunity cost. An owner's unpaid time is an example. (*Chapter XI*)

Income elasticity. The percentage change in quantity demanded divided by the percentage change in consumers' income. A positive result means more income raises the quantity demanded. Example: income up 10% and fiction up 15% give 1.5. (*Chapter V*)

Indifference curve. A line on a graph that joins every combination of two goods giving a consumer the same satisfaction. Because the satisfaction is equal along the line, she would not mind which combination she received. Example: 4 notebooks with 14 pens and 6 notebooks with 8 pens may lie on one curve. (*Chapter VII*)

Indifference map. A set of indifference curves drawn on one graph. A curve further from the corner where the axes meet shows greater satisfaction, so the map ranks every combination of the two goods. Example: 6 notebooks with 14 pens lies on a higher curve than 6 notebooks with 8 pens. (*Chapter VII*)

Industry. All the firms that produce the same kind of good. The quantity supplied by the industry is the sum of what its firms supply. All the firms that make notebooks, for example, form the notebook industry. (*Chapter XIII*)

Inelastic demand. Demand for which the percentage change in quantity is smaller than the percentage change in price, so price elasticity of demand is below 1. A price cut lowers total revenue. Salt is a likely example. (*Chapter V*)

Interest. The payment a borrower makes to a lender for the use of money over time. It is stated as a yearly rate, which is the year's interest divided by the sum borrowed. Tk 100 paid on a loan of Tk 1,000 for a year is a rate of 10 per cent. (*Chapter XVI*)

Isocost line. A line showing all combinations of two inputs that cost the same total amount. Its slope is the ratio of the two input prices. With a wage of Tk 600 and a machine rent of Tk 400, every

6.t2 The toll is only what users pay. Many would have paid more, and the gap between what they would pay and the Tk 20 is their surplus. The benefit is the toll plus this surplus, so it is larger than the tolls collected.

6.1 Consumption is the use of goods and services by people to satisfy their wants. Example: drinking water, or travelling by bus.

6.2 Price is what the consumer must pay, Tk 40 in the notebook case. Willingness to pay is the most the consumer would pay, for instance Tk 60 for the first customer. The gap, Tk 20, is that customer's surplus.

6.3 At Tk 50: $(60 - 50) + (50 - 50) = \text{Tk } 10$. At Tk 20: $40 + 30 + 20 + 10 + 0 = \text{Tk } 100$.

6.4 At Tk 16: $(30 - 16) + (22 - 16) + (16 - 16) = 14 + 6 + 0 = \text{Tk } 20$. At Tk 10: $20 + 12 + 6 + 0 = \text{Tk } 38$. The rise is $38 - 20 = \text{Tk } 18$.

6.5 Willingness to pay falls with each unit, because marginal utility falls. The last unit bought has a willingness to pay close to the price, so the gap between them is close to zero.

CHAPTER VII · INDIFFERENCE CURVE ANALYSIS

7.p1 No. Q has the same notebooks as P but 2 more pens, so Q is better. Two combinations on one curve must please equally.

7.p2 As she gives up pens, those that remain become more precious to her, and as notebooks pile up each extra one is worth less. So she parts with fewer pens for each extra notebook.

7.t1 The old combination now costs $6 \times \text{Tk } 40 + 8 \times \text{Tk } 10 = \text{Tk } 240 + \text{Tk } 80 = \text{Tk } 320$, which is less than Tk 400, so she can still buy it and has Tk 80 left. She can therefore reach a higher curve, so she is better off. The exact new best point depends on her curves.

7.t2 Suppose two curves cross at X. Take A on one curve to the right of X, and B on the other curve directly above A, with the same notebooks and more pens. B is better than A. But A and B each please equally with X, so they would be equal. The two statements clash, so the curves cannot cross.

7.1 $\text{Tk } 200 \div \text{Tk } 40 = 5$ notebooks, or $\text{Tk } 200 \div \text{Tk } 20 = 10$ pens. Two others: 2 notebooks and 6 pens ($80 + 120 = 200$); 3 notebooks and 4 pens ($120 + 80 = 200$).

7.2 Pens given up: $8 - 5 = 3$. Notebooks gained: $8 - 6 = 2$. Rate = $3 \div 2 = 1.5$ pens per notebook. It is smaller than 3, because she now holds fewer pens and more notebooks, so she gives up fewer pens for each notebook.

7.3 Notebooks: $450 \div 50 = 9$. Pens: $450 \div 30 = 15$. Steepness: $50 \div 30 = 1.67$ pens per notebook. Check: $9 \times 50 = \text{Tk } 450$.

7.4 Notebooks: $400 \div 50 = 8$, down from 10. Pens: $400 \div 20 = 20$, unchanged. The line turns inwards around the pen point and becomes steeper: $50 \div 20 = 2.5$ pens per notebook.

7.5 No. The price ratio is 2, but her rate is 4: she would give up 4 pens for one more notebook while the market asks only 2. She gains by buying more notebooks and fewer pens, moving towards 6 notebooks and 8 pens.

machines saved divided by the workers added along an isoquant, and it falls as workers replace machines. At the least-cost combination it equals the price ratio. (*Chapter XII*)

Marginal revenue. The extra revenue from selling one more unit of output. For a price maker it is smaller than the price, because a cut in price applies to every unit sold. Example: selling 10 more copies, from 20 to 30, adds Tk 300, or Tk 30 a copy. (*Chapter XIV*)

Marginal revenue product. The extra revenue a firm gains from employing one more unit of labour. A firm hires while it is at least equal to the wage. Example: an assistant who adds Tk 40,000 a month. (*Chapter XV*)

Marginal utility. The extra satisfaction a person gets from consuming one more unit of a good. It is the height of the last step, not the height of the whole staircase. If the third cup adds 5 utils, its marginal utility is 5 utils. (*Chapter III*)

Market structure. The set of features of a market, such as the number of sellers, the kind of good, the ease of entry and the information buyers hold, that shape how firms behave. A row of identical potato stalls and a lone village pharmacy have different structures. (*Chapter XIII*)

Microeconomics. The branch of economics that studies choices made by individual consumers and firms. Its unit is one consumer, one firm or one market. A question about how one bookshop chooses its price belongs to microeconomics. (*Chapter I*)

Monopolistic competition. A market with many sellers of similar but not identical goods, and free entry. Each seller has a small power over its price, but new entrants take away extra profit over time. Example: a street of tea stalls. (*Chapter XIV*)

Monopoly. A market in which one seller supplies a good with no close substitute. Entry is blocked by barriers such as sole ownership of a resource, an exclusive legal right or a large cost advantage. Example: the only ferry across a river. (*Chapter XIV*)

Nominal wage. A wage measured in money, without allowing for changes in prices. It is the figure on the pay slip. Example: Tk 35,000 a month. (*Chapter XV*)

Normal good. A good for which demand rises when buyers' incomes rise. The word normal does not mean usual. If families with more money buy more notebooks, the notebook is a normal good. (*Chapter IV*)

Normal profit. The least profit a firm must earn to stay in business, covering the owner's implicit costs, such as the value of the owner's own time. A baker who could earn a wage in a factory must earn at least that from her bakery. (*Chapter XIII*)

Normative economics. Statements about what ought to be, which depend on values and cannot be tested by facts alone. Words such as should and ought often signal them. Textbooks should be cheaper is a normative statement. (*Chapter I*)

Oligopoly. A market dominated by a few large sellers, each aware of what the others do. Entry is hard, and each firm's best choice

2.5 Shelving gives $30 \times 20 = 600$ books; packing gives $10 \times 10 = 100$ parcels. All 40 hours are used, and at 600 books the curve allows 100 parcels, so the point lies on the curve.

2.6 First step: 10 buses lost for 5 cars, 2 per car. Second step: 20 buses lost for 5 cars, 4 per car. The opportunity cost rises.

2.7 Resources differ in their suitability, so moving them costs more and more. More resources, or better methods, would shift the curve outward.

CHAPTER III · UTILITY AND DIMINISHING MARGINAL UTILITY

3.p1 Higher. The second cup added some satisfaction, so the total rose, even though the second cup pleased her less than the first.

3.p2 2 utils (25 minus 23). The marginal utility column gives it directly.

3.t1 By the fifth cup her want for tea has been met by the first four cups. She is no longer cold or thirsty, so a further cup adds nothing.

3.t2 Total utility = $25 + (-3) = 22$ utils. She should refuse the sixth cup, because it lowers her total satisfaction.

3.1 Utility is the satisfaction a person gets from using a good or service. Example: the relief of a cold drink on a hot afternoon.

3.2 Totals: 50, then $50 + 35 = 85$, then $85 + 20 = 105$, then $105 + 5 = 110$ utils.

3.3 Marginal utilities: 20, then $36 - 20 = 16$, $48 - 36 = 12$, $54 - 48 = 6$, $56 - 54 = 2$ utils.

3.4 As a person consumes more units of a good, the satisfaction added by each extra unit falls. It holds when the units are alike, tastes stay steady and consumption has no long gaps.

3.5 The marginal utility of the 5th book is 0 utils. Its opportunity cost is the other good the money could have bought, which would have given some satisfaction, so the student gives up something for nothing.

3.6 Falling marginal utility means each addition is smaller, not that the total shrinks. The total keeps rising while marginal utility is above zero.

CHAPTER IV · DEMAND AND THE DEMAND CURVE

4.p1 50 notebooks. A lower price (Tk 30 against Tk 50) brings a larger quantity demanded, which is the law of demand.

4.p2 A movement along the curve, because the price of the good itself changed, and the quantity demanded falls from 40 to 30.

4.t1 Buyers may take a high price as a sign of quality or rank, or may expect the price to rise further. Their tastes or expectations then change with the price, so other conditions are not equal. A ring is not a very basic good, so it is not a Giffen good.

4.t2 The fare rise is a movement along the bus demand curve, with a smaller quantity demanded. Cheaper petrol makes car travel cheaper, and cars are a substitute for buses, so demand for bus travel shifts to the left.

profit says owners are paid more on average for facing this chance. (*Chapter XVI*)

Scarcity. The condition that resources are limited while wants are unlimited, so choices must be made. It applies to rich and poor alike, and it is the reason economics exists. Example: a family cannot buy everything on its list. (*Chapter II*)

Short run. A period so short that at least one input, such as a machine, cannot be changed. It is not a fixed number of days; it is whatever period leaves an input fixed. Example: a baker can hire helpers but cannot install a new oven. (*Chapter X*)

Snob effect. A fall in demand for a good because many other people now buy it. A buyer who values rarity leaves a good that has become common, as when a once-rare scarf is worn by everyone. (*Chapter IV*)

Substitute goods. Goods that can replace each other, so a price rise for one raises demand for the other. If a rival notebook becomes dearer, some buyers move to ours, and demand for our notebook shifts to the right. (*Chapter IV*)

Supply. The quantities of a good that sellers are willing and able to sell at each price in a period. Both the stock and the wish to sell are needed. One entry is a quantity supplied; the whole list is supply. For the notebook: 40 at Tk 40, 50 at Tk 50, and so on. (*Chapter VIII*)

Supply curve. A line on a graph showing the quantity supplied at each price. Price is on the vertical axis and quantity on the horizontal axis. A point on the line means that at this price sellers offer this quantity. Usually the line slopes upward. (*Chapter VIII*)

Supply schedule. A table showing the quantity supplied at each price. Each row pairs one price with one quantity. The wholesaler's table lists Tk 20 with 20 notebooks, Tk 30 with 30, and continues up to Tk 60 with 60. (*Chapter VIII*)

Total cost. The whole cost of producing a given output, found by adding fixed cost and variable cost. At zero output it equals fixed cost. A workshop with fixed cost of Tk 600 and variable cost of Tk 720 has total cost of Tk 1,320. (*Chapter XI*)

Total product. The whole quantity of output produced with a given amount of inputs. Example: the binding workshop with 4 workers has a total product of 44. (*Chapter X*)

Total revenue. Price multiplied by quantity sold; the money a firm receives from its sales. Example: 40 notebooks sold at Tk 40 give total revenue of Tk 1,600. The total revenue test uses it to judge the effect of a price change. (*Chapter V*)

Total utility. The whole satisfaction a person gets from all the units of a good consumed in a period. It is found by adding the marginal utility of each unit. If three cups gave 10, 8 and 5 utils, total utility is 23 utils. (*Chapter III*)

Trade-off. Giving up some of one thing in order to have more of another. It is the exchange itself, while opportunity cost measures what is given up. Example: planting more jute and less rice.

Util. An imaginary unit used to count satisfaction in economic examples. A util is not a real measurement; it lets us reason exactly about more and less for one person. A cup worth 10 utils pleases more than a cup worth 8 utils. (*Chapter III*)

Utility. The satisfaction a person gets from using a good or service. It is a feeling, not a judgement of whether the good is wise or healthy. A cold drink on a hot day gives utility to the person who drinks it. (*Chapter III*)

Variable cost. A cost that changes with the quantity produced. It is zero when nothing is made and rises as output rises. Paper, thread, glue and the wages of extra workers are examples in a binding workshop. (*Chapter XI*)

Wage. The payment to labour for its work over a period. It may be stated by the hour, the day or the month. Example: Tk 35,000 a month for a shop assistant. (*Chapter XV*)

Want. Something a person would like to have, however little or much the person can pay for it. A want may be a good, a service or a condition. Example: a bicycle, a haircut or a free afternoon. (*Chapter II*)

Willingness to pay. The highest price a buyer would pay for one more unit of a good. It is a limit held in the mind, not a price on a tag. It falls as a consumer buys more units and it differs between consumers. A consumer buys a unit only when the price is at or below this limit. (*Chapter VI*)

How do buyers and sellers decide, and what do their choices do to prices? This treatise answers from the beginning. It sets out the nature of economic choice, then follows it through demand and supply, the theory of the consumer, production and cost, and the behaviour of firms in competition and monopoly. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.